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ARTICLE

The detection floor bounds tolerance endpoints in *Mycobacterium tuberculosis*

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Abstract

A culture that grows nothing after antibiotics has never proved sterility, so tolerance is scored by how far the count fell: the minimum duration for killing. That endpoint is a fraction of the starting count, so the inoculum should cancel. It does in the definition, and fails once killing reaches the assay's floor. Five deposits are reanalysed; one experiment tests this.

Two boundaries follow from the definitions. For a floor L, a q-log endpoint is visible only above $L{\cdot}10^q$; above L/c_1, with c_1 the lowest class threshold, a floor-level reading fits only the lowest class. Reaching the floor is the drug's doing; the label it permits is not.

Of 217 *Mycobacterium tuberculosis* isolates classified for rifampicin tolerance, 33 began too close to the floor to show the 99.99 per cent reduction used, and all 33 are recorded as failing. Of eighteen read at the floor, twelve admit one class, six admit two. The label tracks growth state, not isoniazid resistance; most of that runs through a ten-fold lower inoculum — the denominator it is cut on.

Prospectively, *Escherichia coli* ATCC 25922 at the standard inoculum could report 3.71–3.93 logs at 10 µL and 4.88–5.05 at 100 µL; at a cut of 10^{-3}, six of 54 sample-times took two labels from one culture, the pipette alone deciding. Letting the plated volume vary across the permitted range moves demonstrable depth by 1.60 log10, as exact arithmetic. A tolerance call reported without its starting density and assay floor cannot be interpreted.

Keywords: time-kill assay; limit of quantification; minimum duration for killing; antibiotic tolerance; colony counting; censored data; *Mycobacterium tuberculosis*

Introduction

Tuberculosis treatment runs for four to six months (1). Shortening it means finding drugs that kill *Mycobacterium tuberculosis* faster, and that choice is made in a flask: the compounds that could shorten a regimen are picked out of in vitro killing experiments.

The experiment is simple. Grow the organism, add the drug, and at set times take a sample, dilute it, spread it on a plate, and count the colonies. Each colony stands for one bacterium that was alive when it was plated. The counts over time are the killing curve.

Every clinical microbiologist already knows the trap in that last step. A culture taken after antibiotics have started is not evidence of sterility. It is why blood cultures are drawn before the first dose. A drug suppresses growth, injures cells sublethally, carries over into the medium, and drives the population beneath what the plate can see; the plate then reports absence, and absence is not what happened. The rule is old, it is correct, and nobody disputes it.

The plate also has a hard floor, and the pipette sets it. The smallest positive result is one colony, and what one colony means depends on how much you spread. Plate 100 µL and one colony is 10 bacteria per mL. That is the floor. Anything below it comes back as no growth, which is not the same as nothing there. Write L for that floor and *No* for the starting density.

Tuberculosis makes the old rule expensive, because the phenotype invoked to explain why treatment takes so long is itself defined by a duration. That phenotype is tolerance: the capacity of a genetically susceptible population to survive an exposure that should kill it. These bacteria are not resistant. The drug still works on them. They just take a long time to die, and that time is the measurement. A phenotype defined by a duration is defined by exactly the observation the old rule warns about.

The field's response was not to ignore the warning but to engineer around it. Rather than asking whether a culture went negative, the modern definition asks how many logs the population fell: the minimum duration for killing, MDK, is the time to a specified fractional reduction — 90, 99 or 99.99 per cent — and it is proposed as the tolerance counterpart to the minimum inhibitory concentration (2, 3). The choice of a *fraction* is the whole point. A ratio to the starting population is scale-free: if the count falls in a straight line on a log scale at rate b, the time to a q-log reduction is q/b, and the starting density cancels. By construction, MDK cannot be contaminated by how much culture went into the tube. That is the property it was built to have.

The property holds in the definition. It does not hold in the measurement, and the gap between the two is what this paper is about. To show a four-log kill you have to be able to see four logs down. A q-log reduction can be estimated only if q logs are visible, and what is visible is bounded below by the floor. So an isolate has

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headroom  h = log10(N₀ / L)
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and no experiment can demonstrate a reduction deeper than h, however completely the drug worked. Where $h < q$ the endpoint is unreachable before the drug is added. That is the first boundary, and we call it reachability.

It gets worse when the last count lands on the floor. The fraction written down is then L/N_0 — the floor divided by the starting density. It is not a measurement of how many survived. The drug has dropped out of it, and what is left is the pipette and the inoculum. The metric designed to be inoculum-independent becomes a pure inoculum readout precisely at the depth where tolerance is scored. That is the second boundary, and we call it identifiability.

The two failures are different in kind, and this paper keeps them apart. Whether an endpoint is reachable at all is settled by the assay geometry before the drug is added, since h depends only on the starting culture and the floor. Whether a reading that has reached the floor can be classified is settled by the geometry too, but only once the drug has driven the population down to it: reaching the floor takes a reduction of the isolate's entire headroom, and that is the drug's doing.

The bound is already in the papers that defined the assays. Vijay and colleagues, introducing the most-probable-number MDK for rifampicin — a count read off a dilution series rather than off colonies on a plate — treated an MDK99.99 longer than ten days as high tolerance beyond the **time** window of the assay, not as a statement about the count floor (4). In the 217-isolate classification that follows from that assay they report that MDK99.99 at 15 days of recovery could be calculated for only 22 of 209 isolates and that the rest lay, in their words, "beyond the assay limits" (5). The ERA4TB consortium requires below- and above-quantification-limit flags together with the limit of quantification, and notes that pharmacometric models can use those flags (6); the same six-laboratory comparison then excluded both from its numerical analysis. Their 2025 protocol states the geometry of the pipette explicitly: a 2.5 µL drop has a limit of detection above 2.6 log10 CFU/mL, and a lower limit requires a larger plated volume (7). Independently, a 2025 pharmacokinetic-pharmacodynamic analysis of hollow-fibre CFU series showed that censoring counts below 10 CFU — recording them only as "below the floor", with the true value unknown — biases regimen ranking (8), and a within-host tuberculosis tolerance study chose a shallower MDK threshold specifically so that more isolates could enter the analysis (9).

The closest statement of the bound is in the framework's own methods paper, and we claim no priority over it. Brauner and colleagues, presenting a direct measurement of tolerance that deliberately avoids time-kill curves, scale the inoculum to the endpoint being measured: a hundred bacteria per well to determine MDK99, a thousand for MDK99.9, and so on (3). That is the reachability inequality of this paper, with the floor idealised at one cell per well and a grew-or-did-not-grow readout in place of a count. What has not been done is to carry the rule back into plate-count time-kill, where the floor is not one cell but a concentration fixed by the plated volume — 10 to 400 CFU/mL across the deposits reanalysed here — and must therefore be measured sample by sample rather than assumed; nor to ask what the published record looks like where the rule was not applied.

The requirement is older still, and it is quantitative. NCCLS M26-A ties the plated volume to the endpoint: after the defined 99.9 per cent killing, at least ten colonies must remain to be counted. It separately requires the smallest accurately detectable count to be established by serial dilution of a known inoculum (10). What has happened since is that the endpoint moved and the rule did not travel with it. The persistence field's own consensus guideline makes the time-kill assay the starting point for identifying persistence and catalogues at length what can go wrong with it — resistant mutants, heteroresistance, drug degradation, adhesion to the vessel, washing and recovery conditions, delayed colony appearance, culture density, inoculum history — and raises neither the limit of detection, nor the plated volume, nor how deep a reduction the assay can report (11).

None of these converts a floor-level reading into the classification rule it becomes. When the last count sits at L, the recorded fraction is L/N_0, and the low, medium or high label is then a cut on starting density. That is the gap this paper fills. Two further failures have to be kept distinct from it. A duration ceiling — the assay stopped looking — is not a count floor, which is the plate being unable to see. And recording a below-limit flag is not the same as scoring a duration from the points so recorded.

This is testable rather than arguable, because one published deposit assigns the tolerance phenotype itself. Vijay and colleagues (5, 12) scored 217 clinical *Mycobacterium tuberculosis* isolates as having low, medium or high tolerance to rifampicin, and deposited alongside each call the starting most probable number, the growth rate, the isoniazid susceptibility and the killing readings the call was built from. The classification, the inputs to it, and the assay geometry that bounds it are all in the same file.

This study asks how deep a log-reduction tolerance endpoint can be measured at all, and whether real isolates have been assigned tolerance phenotypes from outside that range. Two boundaries on the starting density follow from the definitions — one deciding whether an endpoint is reachable, the other whether a floor-level reading identifies a class — and both are computed from quantities a time-kill protocol already records. Neither asks a laboratory to measure anything new.

We then test the hypothesis that a deep log-reduction endpoint is inoculum-dependent through what can be observed, even though it is inoculum-independent in its definition, using the classification and the assay geometry of 217 clinical isolates. We then ask what the resulting phenotype tracks: drug susceptibility, or the physiological state of the culture. Two further deposits establish that the same arithmetic governs killing experiments where inoculum is controlled by protocol rather than by clinical accident: a six-laboratory consortium exercise distributing one strain under one written protocol (6, 13), and a concentration-by-time grid in the same organism (14, 15). From the first we also ask whether a faster-killing flask reliably crosses the floor sooner. From the second, whether a late endpoint can erase a dose difference that an early one resolves. A fourth deposit (16, 17), held out until every boundary was fixed, tests whether the boundaries locate the same depth in an experiment they were not built from. Finally we ask whether the concentration and duration axes are separate in these files, which is the premise the framework defining them rests on. All five deposits are published and openly licensed, and none was generated for this study (Table 1). The two boundaries are then tested forward rather than backward, in a prospective experiment built to break them, with every prediction written down before a plate was counted.

If the hypothesis holds, a tolerance call reported without its starting density and its assay floor cannot be interpreted, and the phenotypes assigned in its name are, in part, a record of how the assay was set up.

Box 1. Four numbers that score an MDK. *No* is the starting density. *L* is the smallest positive count the method can report — one colony in a plated volume *v* µL is $L = 1\,000/v$ per mL, and an MPN series uses the lowest table rung (18, 19). Headroom is $h = \log10(No/L)$. A *q*-log endpoint is legal only if $h \geq q$, which is the same as seeding at or above $L \cdot 10\text{^}q$. If the last reading sits at *L*, the recorded fraction is L/No: an upper bound, not a measurement.

Culture	N_0 (per mL)	*L* (per mL)	Headroom	Legal endpoints	If the last reading is at *L*
Thin clinical MPN	23 000	23	3.00	90%, 99%, 99.9%; not 99.99%	Fraction 10^{-3}. Low and medium both compatible: the rule cannot choose.
Adequate clinical MPN	230 000	23	4.00	all four, including 99.99%	Fraction 10^{-4}. Only low is compatible.
100 µL plate	10^5	10	4.00	through 99.99%; not a 5-log call	Fraction 10^{-4}. A 5-log MDK needs $N_0 \geq 10^6$.
10 µL drop, same N_0	10^5	100	3.00	through 99.9%; not 99.99%	Fraction 10^{-3}. The pipette, not the isolate, removed one log.

The first two rows are the two starting densities that recur in the clinical deposit analysed below, and Section 2 reports what became of the isolates that began at each. The last two are why a written protocol does not standardise measurable depth: the plated volume sets *L*. To make both a four-log endpoint and a low class identifiable against *L* = 23 per mL, seed above $L \cdot \max(10^4, 1/c1) = 230\,000$ per mL, or choose a shallower endpoint.

Two of the claims this arithmetic supports are not testable and should not be read as though they were. If the last reading sits at the floor, the recorded reduction cannot exceed h, because a deeper one would require a count below L. And the recorded fraction is then L/N_0, a monotone decreasing function of N_0, so its rank correlation with N_0 is −1 whatever the numbers are, including random ones. Both follow from the definition of a censored reading, and neither is evidence about a drug.

What is testable is what happens when one sample is plated at two volumes under a fixed class threshold. If both readings sit at their own floors the labels are L_1/N_0 and L_2/N_0, and they differ only when those two fractions fall on opposite sides of the cut — which is again arithmetic. The empirical quantity is how often a real experiment lands in that window. It could be none of the time.

Results

1. The plate's floor sets how far down killing can be seen

The most probable number (MPN) readings can only take the discrete values of an MPN table (18, 19) — a fixed ladder of rungs, not a continuous scale. The day-5 column does not taper towards zero. It stops. Eighteen isolates sit at exactly 23 per mL in the 15-day panel and six in the 60-day panel, and nothing in the file lies below that (Fig. 1A). That is a floor, not a tail, and it is treated as one throughout.

The deposit never states its limit of quantification (12), so the floor has to be inferred, and the inference is quantified rather than asserted. A posterior over the MPN rungs at or below the lowest observed value — a probability spread across which rung is the real floor — places 95 per cent support on 9.2 to 23 per mL, a span of 0.40 log10. That is narrow enough that the class labels of Section 3 are computed rather than refused (Table S1). Where a deposit gives no such evidence, the labels are refused instead of reported.

The deposit carries a second classification, scored at day 2, and it sits on a different floor. The day-2 column bottoms out at 230 per mL with eighteen readings resting there, so L = 230 there. The day-2 class thresholds are not stated in the source either, and they are recovered the same way as the day-5 ones: cuts at 10^{-2} and 10^{-1} reproduce all 203 usable day-2 classes with no disagreement, one decade shallower than at day 5. Floor and lowest threshold both shift by a factor of ten, so the identifiability boundary does not move: $N\_id = 230/10^{-2}$ = 23 000 per mL, the same value as at day 5. The reachability boundary does move, by a decade, and the consequence is large. Against the day-2 floor, 121 of 203 isolates have too little room for a four-log reduction; against the day-5 floor, 31 of those 203 isolates do. Eighteen day-2 readings rest on their floor, eleven with a single compatible class and seven with more than one. At 60 days no pair of decade cuts reproduces the day-2 classes (109 of 197), so that rule is not a decade threshold on the recorded fraction, and no day-2 accounting is offered for it.

Each isolate therefore carries a fixed budget of killing the assay can see. At 15 days of prior culture the starting densities span 3.36 to 7.79 log10, so headroom — the distance from the starting density down to the floor — spans 2.00 to 6.42 log10 (Fig. 1B, Table 2). Both spans are the same 4.42 log10, and the fold figures quoted throughout are ten raised to the unrounded difference rather than to the displayed one. Against that budget the three deposited endpoints behave very differently. Every isolate has the room for a 90 or a 99 per cent reduction: 0 of 217 fall short in both cases. **Thirty-three of 217 isolates — 15.2 per cent — lack the headroom for the 99.99 per cent endpoint**, which is to say that a four-log reduction was unobservable for them before rifampicin was added. Two of those 33 carry a fourth label that the deposit writes literally as "MDR", conventionally multidrug-resistant. It is unordered with respect to the other three, so no ordered analysis can place it (Table S2); over the 203 isolates with an ordered label the same count is the 31 quoted above.

All 33 are recorded as not having reached the endpoint — they sit at the **assay ceiling**. That is a census of the affected isolates rather than an estimate, and it is the load-bearing observation. The ceiling is a different censoring mechanism from the floor and should not be read as its mirror. The floor censors the count: anything below *L* is not reported as a count. The ceiling censors time: the killing assay runs six days, so an isolate that has not reached its endpoint by then is recorded at the last day rather than at the day it would have reached. One sets how deep the assay can see, the other how long it looks. Among the 184 isolates that did have four logs of headroom, 162 are also at the ceiling, 88.0 per cent.

No p-value is attached to that contrast, and the reason is worth saying out loud rather than hiding in a caveat. An isolate with less than four logs of headroom cannot record a four-log reduction. The 0 of 33 cell is fixed by arithmetic, not by biology, so the null of independence between ceiling status and headroom is false before any data are seen. A test of an impossible null still returns a number, and that number means nothing. The census is the finding: every isolate that could not reach the endpoint is recorded as not having reached it, and 88.0 per cent of those that could are recorded the same way. The deepest endpoint is therefore the only one at risk, and it is the one the tolerance classification is built on.

2. The same floor reading was given different tolerance labels

The deposited tolerance level is a cut-off on the recorded surviving fraction, with no overlap between classes. The thresholds themselves are not deposited. They are recovered here, and the data pin them only to the gaps between classes: at day 5 and 15 days of prior culture, low tolerance covers fractions below 10^{-3}, medium covers 10^{-3} to 10^{-2}, and high exceeds 10^{-2}. Those cuts reproduce every usable class in the file, 203 of 203 (Table S3). Usable means the 203 of 217 isolates whose label is one of the three ordered classes; the other 14 carry the fourth, unordered category, the "MDR" label above, that no ordered analysis can place (Table S2). So what follows is an argument about the fraction the assay recorded, not about an independent clinical judgement.

Eighteen isolates ended at the floor. They share one reported floor-level observation, but their true final counts are unknown below the floor, so the assay cannot tell their surviving burdens apart down there. That is not the same as saying they were killed to the same degree, and the difference matters. Their starting densities span

265-fold, from 23 000 to 6.1×10^6 per mL, so the same terminal reading implies a different range of compatible reductions in each. An isolate that began at 23 000 has demonstrated at least a 3.00-log reduction; one that began at 6.1×10^6 at least 5.42 logs. In either case the true reduction may be anything from that bound down to complete kill. What the deposit records instead is a single number per isolate, L/No, and those recorded fractions span 265-fold, from 3.8×10^{-6} to 1.0×10^{-3} — exactly the span of the starting densities (Fig. 1C), because that is what the ratio is. They are upper bounds on survival, not measurements of it.

The labels those eighteen received differ. Six isolates that began at 23 000 per mL received a fraction of 10^{-3} and were classified **medium** tolerance; twelve that began at 230 000 or above received 10^{-4} or less and were classified **low**. A single cut on the starting density alone reproduces all eighteen (Table 3). That is a statement about the classification step and not about the killing: once a reading is censored at the floor, the starting density fixes which label the published rule is able to return.

The question this raises is not whether those labels are wrong but which labels the published rule could have returned once the reading came back at the floor. A censored reading places the true count somewhere between zero and L. Sweeping the true count across that range and applying the deposited thresholds gives the set of classes compatible with the observation. For the twelve isolates that began at 230 000 per mL or above, every count the assay admits yields a fraction below 10^{-3}, so **low** is the only class compatible with a floor-level reading. For the six that began at 23 000, the compatible set spans the low–medium threshold and the rule cannot choose between two classes.

Three questions have to be kept apart here, and the rest of this paper keeps them apart. Whether a culture reaches the floor at all is a question about the drug: rifampicin has to reduce the population by that isolate's entire headroom for it to happen, and the starting density does not by itself decide it. Which label the rule assigns once the reading is censored is a question about arithmetic: there the starting density determines the compatible set, and for twelve of these eighteen that set has one member. How far the population actually fell is a question the assay leaves open, since the true count lies anywhere below L. The first is biology, the second is bookkeeping, and the third is unmeasured. Conflating them is the error this paper is about.

So for twelve of the eighteen the recorded label carries no information beyond the fact that the reading was censored, and for six it is not determinate at all. None of the eighteen is a measurement of the surviving fraction.

Sorting every call in the panel this way separates the labels the assay measured from the labels the censoring rule fixed (Table 3). At 15 days of prior culture, 185 of 203 calls rest on a reading above the floor and are measured; for 12 the reading is censored and only one class is compatible with it; for 6 the reading is censored and more than one class is compatible. At 60 days the counts are 191, 6 and none, because the cultures are denser and few readings reach the floor.

The larger effect is at the deeper endpoint, and it falls unevenly on the groups a study would compare. The 99.99 per cent endpoint is unreachable for 22 of 84 isoniazid-resistant isolates, 26.2 per cent, against 9 of 119 susceptible ones, 7.6 per cent (Fisher exact $p = 0.00056$ (20)); the remaining two of the 33 are MDR and fall outside that contrast. Their median headroom differs by a full log, 4 against 5. A comparison of tolerance between those groups is therefore in part a comparison of how well each group could be measured, which is the disposition the previous section quantifies.

3. The affected isolates are named before the drug is added

Sections 1 and 2 counted isolates. The two boundaries derived in the Methods do more than that. Given only the floor, the class thresholds and each isolate's starting density, they say in advance which isolates are affected, before any day-5 reading is used.

For this assay $N\_reach = 23 \times 10^4 = 230\,000$ per mL and $N\_id = 23/10^{-3} = 23\,000$ per mL. Applied to the 15-day panel, the first predicts 33 isolates unable to reach the 99.99 per cent endpoint. The second predicts that of the eighteen isolates whose reading rests on the floor, twelve have only the lowest class compatible with that reading and six have more than one. Those are the three counts Sections 1 and 2 report.

Neither boundary predicts which isolates reach the floor; that depends on what rifampicin does. Both predict what a reading at the floor can be made to mean once it happens, and there the agreement with the deposit is exact isolate by isolate: the set the algebra says has a single compatible class and the set the deposit records in the lowest class are the same set.

That agreement is an identity rather than a passed test, and saying so is the honest way to report it. Once the label is a cut on the recorded fraction (Table S3) and every floor reading has a numerator of exactly L, the class of a floored isolate is a function of N_0 alone, so the algebra cannot disagree with the deposit. What could have failed, and did not, is the premise: that the three classes separate cleanly on the recorded fraction at all, in 203 of 203 calls.

How finely the agreement resolves also has to be stated. Among the eighteen the starting density takes only five values — 23 000 for six isolates, then 230 000 for eight, and 610 000, 2.3×10^6 and 6.1×10^6 for one, one and two — and **no isolate lies strictly between N_id and N_reach**. Any cut placed anywhere in that decade reproduces the same twelve-six split, so the split locates the boundary only to within an order of magnitude and is not by itself a sharp test of where it sits.

What is sharp is the boundary case, and it is the part of this section that discriminates. Six isolates sit at exactly 23 000 per mL, which is N_id itself. For them L/N_0 equals c_1 exactly, the interval $[0, L/N_0]$ touches the threshold from below and spans two classes, and the strict inequality fails. Those six are precisely the six the deposit records as medium rather than low. A rule written with the inequality the other way would have called all six low and been wrong six times out of six. That is a test the arithmetic could have failed and did not, and it is what licenses using the boundary in advance: given a floor, a set of thresholds and a starting density, the affected isolates can be named before any reading is taken.

The empirical findings are therefore consequences of the definitions rather than properties of this deposit, and the claim is a prediction that any dataset reporting a starting density, a floor and a threshold classification can be checked against.

4. The label tracks how fast an isolate grows, and its link to resistance cannot be separated from the inoculum

We asked what the deposited tolerance label goes with. There are two candidate predictors, and each is tested at two culture ages and at two endpoint depths, which is eight tests in all. Eight tests on one question will throw up a false positive on their own, so we correct them together as one family. Two survive (Table S4).

The label tracks the **growth rate** of the isolate. Slower growth goes with a higher tolerance class, and it survives adjustment for starting density (OR 1.096 per day of time to OD 0.4, 95 per cent CI 1.032 to 1.164, $p = 0.0030$, $n = 202$; unadjusted OR 1.126, 1.062 to 1.193). The label also tracks **isoniazid resistance** — until the starting density enters the model. On its own, a resistant isolate has 2.32 times the odds of a higher tolerance class (95 per cent CI 1.30 to 4.12, $p = 0.0042$). Adjusted for starting density the odds ratio falls to 1.31 (0.67 to 2.57, $p = 0.42$), and the interval no longer excludes one. We call that attenuation, not elimination. Starting density is the strongest term in the file: every ten-fold rise in the starting most probable number halves the odds of a higher tolerance class (OR 0.480, 0.340 to 0.677, $p = 2.9 \times 10^{-5}$).

Why the adjustment does that is measurable. Isoniazid-resistant isolates go into this assay at 5.36 log10 against 6.36 for susceptible isolates, ten-fold thinner (Mann–Whitney $p = 9.1 \times 10^{-14}$) (21), and 26.2 per cent of them lack the headroom for the deepest endpoint against 7.6 per cent of susceptible isolates (Section 2). They are one log short of observable killing before the experiment begins.

A percentage attenuation is a description, not a quantity a model estimates, so we estimated the path directly. The total effect of resistance on the tolerance class splits into two parts: the part that runs through log10 starting density and the part that does not. On the 203 isolates with a usable class, rather than the 202 the association family retains, the mediated part is +0.166 classes (bootstrap 95 per cent CI +0.067 to +0.279) and the direct part +0.091 (−0.116 to +0.287). About 65 per cent of the association therefore travels through the inoculum, and the direct path covers zero. Restricting to baseline isolates leaves the mediated path intact (+0.159, +0.053 to +0.295), and collapsing the outcome to two levels agrees (Table S5).

Two things bound that estimate, and neither is a formality. It rests on sequential ignorability — that nothing unmeasured drives both the starting density and the class — which no observational file can establish. A residual correlation of about −0.23 between the mediator and outcome errors would nullify the point estimate, and one ordinary measured covariate already moves it about forty per cent of that way. More fundamentally, the step in the middle is the denominator of the ratio the outcome is cut on. The tolerance class is a threshold on N_5/N_0, so a decomposition that routes resistance through N_0 is in part recovering the censoring arithmetic this paper is about rather than a biological pathway from resistance to tolerance. That is not a reason to discard the estimate — it is the quantity a reader wants, because it says how much of the association is carried by the label's own denominator — but it is a reason not to read it as a mechanism. Only a design that fixes the inoculum separates the two.

That last objection can be tested rather than merely conceded, because the isolates it applies to can be named. The recorded fraction is L/No exactly for the eighteen isolates whose day-5 reading sits on the floor, and for no others: in the remaining 185 the numerator was counted and is free to move independently of No. Refit on those 185 alone and the mediated path does not weaken but strengthens, to +0.227 classes (+0.118 to +0.348), while the direct path collapses to +0.008 (−0.197 to +0.208) and the proportion mediated rises from 0.65 to 0.96. Deleting exactly the isolates the criticism is about leaves the effect larger, so it is not an artefact of them. What it remains is an association in observational data, and the caution above about mechanism is unaffected.

These 217 isolates are not 217 independent patients, and the deposit carries no patient key with which to say so in a standard error. What it does carry is the exact substitute: a baseline-only stratum in which every isolate comes from a different patient by construction (Time_point = 0M, n = 167 at 15 days).

In that stratum the two surviving associations part company. Growth holds and still survives correction (OR 1.116, 1.040 to 1.198, $p = 0.0024$). Resistance does not: OR 2.11 (1.07 to 4.16, $p = 0.031$). It ranks second in the family of eight, so under Benjamini–Hochberg it must beat a critical value of 0.0125, and it does not. The resistance association therefore leans on isolates that are not independent of one another, and it is already the association that starting density explains away. We report it as the weaker of the two on both counts.

The two surviving associations sit in the 15-day panel, which is where the mechanism places them. By 60 days the cultures are 38-fold denser, the spread of starting densities has more than halved, from an interquartile range of 1.00 to 0.42 log10, and the fraction of isolates short of headroom falls from 15.2 to 3.3 per cent. The confound is a property of a thin assay, and it thins out when the assay is not thin. The two panels are columns on the same spreadsheet rows — 210 isolates appear in both — so every comparison between them is within-isolate and is made that way. Pairing sharpens the result rather than softening it: 26 isolates lose their headroom shortfall between panels and none acquires one (exact McNemar $p = 3.0 \times 10^{-8}$ against an unpaired p of 2×10^{-5}) (22), eleven leave the floor and none joins it ($p = 9.8 \times 10^{-4}$), and every isolate whose headroom changes gains (Wilcoxon signed-rank $p = 1.3 \times 10^{-33}$) (23). The interquartile range of starting density falls by 0.58 log10.

It does not vanish, and the ordinal treatment is what shows where it goes. The one-odds-ratio-for-both-cuts assumption holds throughout the 15-day panel but fails at 60 days for starting density at the deepest endpoint (Brant $p = 0.0031$; likelihood-ratio test against a released fit $p = 0.0062$). Let that coefficient differ between the two cuts and the whole effect lands on the upper one: the odds ratio for clearing medium is 0.493 (0.336 to 0.724, $p = 0.0003$) against 0.951 for clearing low (0.647 to 1.396, $p = 0.80$). A multinomial fit, which assumes no ordering at all, agrees (high against low, relative risk ratio 0.59, $p = 0.028$; medium against low, 1.22, $p = 0.39$). In the denser panel the starting density no longer decides who is called low. It still decides who can be called high. A single averaged slope, ordinal or linear, would have reported that as a null.

5. One written protocol does not give every laboratory the same depth to look down

If the effect is a property of assay geometry rather than of clinical sampling, it should appear where one protocol, one strain and one stock are handed out deliberately. Van Wijk and colleagues (2023) ran exactly that exercise, sending a single stock of H37Rv to six blinded laboratories under one written protocol (6, 13). It does.

At ten times the minimum inhibitory concentration of moxifloxacin all six laboratories return a positive kill rate, and the rates span 5.2-fold, from 0.090 to 0.464 log10 CFU/mL per day. Across all 30 laboratory-by-arm cells the Tobit estimate — a fit that treats a below-floor reading as below the floor rather than as a number — and the imputation estimate never differ by more than 0.006 log10 per day (Fig. 2A, Table S6). That agreement says the fit is not an artefact of the arithmetic used to reach it. It does not say the sub-floor assumption is safe: both estimators assume the same normal distribution below the floor, and nothing in this deposit can check that assumption from the inside.

The duration endpoint behaves differently on the same flasks. What it records is not clearance and not sterilisation. It is a **first observed crossing below the assay floor**, and the distinction is not pedantic: most series that cross later read above the floor again, including all five untreated series that cross. The plating volume has to be stated too, because the endpoint depends on it. The time-to-event analysis runs at the 100 µL quadruplicate plating, the most sensitive of the four, and there 29 of 72 treated flasks ever fall below the floor: institutes B and C record crossings in every arm, institute A in three of its four, and institutes D, E and F in none (Fig. S1A). For half the laboratories the first-crossing time is therefore right-censored throughout — the flask never crossed, so all that is known is that the crossing, if it comes, comes after the last visit.

That split is a property of the plating, not of the laboratories. Counting a crossing on **any** of the four platings gives 48 of the same 72 treated flasks, and every laboratory records at least one — D six of its twelve, E one, F ten. The two counts are the same fact seen at two sensitivities, and they are exactly what this paper is about: a duration endpoint is defined against a floor, and the floor is a choice. The analysis keeps the single most sensitive plating so that the endpoint means the same thing everywhere, and reports here what the other choice would have shown.

The two orderings do not correspond. Rank the laboratories by starting density and the three lowest are exactly the three that recorded crossings, the three highest exactly the three that did not, without exception. Rank them by kill rate and there is no such correspondence (Fig. 2B). Institute F has the highest starting density at 6.53 log10 among the treated flasks of this arm, kills faster than four of the other five at 0.223 log10 per day, and records no crossing. Institute E returns the slowest rate of all at 0.090 and also records none.

Starting density separates the flasks that ever crossed from those that never did with an area under the curve of 0.974. That figure is a description, and it is quoted without a p-value on purpose. The comparison looks like 67 flasks, but it is three laboratories against three, and the between-laboratory evidence in this deposit is six clusters throughout; the Limitations set out what that permits and what it forbids. In short: the exact laboratory-level test returns 0.10, which is the smallest value the design can return, and once the treatment arm is held fixed there is no within-laboratory comparison left to fall back on.

The Cox fits are reported for the same reason and with the same restraint. On laboratory alone, institutes D, E and F carry hazard ratios of 0.20, 0.21 and 0.20. Add starting density and those move to 0.34, 0.36 and 0.62, and concordance rises from 0.896 to 0.930 (Fig. S1B). Institute C moves the other way, from 3.0 to 5.3, and it is also the fastest killer, which is how a laboratory that genuinely kills faster ought to behave. No p-value is attached to any of these terms, and the panel that displays them displays that movement rather than testing it. The reason is mechanical. The tested covariate is constant within cluster, and with six laboratories and six parameters the cluster-robust covariance is singular by construction; fitting it anyway does not fail loudly but returns a smaller p than the one it was meant to correct. A coefficient that moves on adjustment says how far starting density and laboratory identity overlap in this design. It is not evidence that one explains the other.

Killing here is not sustained, and this is stronger than a caveat about individual flasks. Take the 64 treated series carrying at least three quantified readings at the most sensitive plating volume. In 48 of them the final step is not a decline, and 44 end more than one log10 above their own lowest reading, with a median rebound of 2.17 log10 and a largest of 5.54. Seventeen of the 32 flasks that fell below the floor at the most sensitive plating — the 29 treated flasks above and three untreated ones — were detectable again at a later visit, 14 of them in treated arms.

Across all four platings, 84 of the 140 series that cross return above the floor, 60 per cent, and the return is quick: a median of four days, with 45 per cent back within three. Those 360 series come from 90 flasks, four platings each, so they are not 360 independent observations; counted by flask, 53 of the 90 contain a crossing series — the 48 treated flasks above and five untreated ones — and 42 of those contain a returning one. A crossing below the assay floor in this deposit is therefore usually a transient rather than an endpoint, which is a further reason it summarises less than the trajectory that produced it. Killing itself is real and dose-dependent: at one times the inhibitory concentration five of six laboratories record net growth under moxifloxacin, between −0.047 and −0.214 log10 per day, and at ten times all six record net decline.

Two consequences follow, and they matter for anyone who reads a duration off such an experiment. The first is that the state below the floor is not absorbing: a series that drops below it does not stay there. Of 2 232 visit-to-visit transitions, 144 go from above to below and 95 go from below to above, so a series sitting below the floor leaves it at the next visit with probability 0.22. The gradient runs with drug pressure — that probability is 1.00 in untreated series, 0.37 to 0.92 at one times MIC and 0.07 to 0.18 at ten times — which says the event is at least partly a property of the plate rather than of the drug. Of the 360 series, only 56, or 15.6 per cent, show the shape a survival model assumes: one crossing that holds (Table S7). Two hundred and twenty never cross at all, 65 cross and return, and 19 oscillate more than once.

The second is that the crossing time is never observed. It lies between the last visit above the floor and the first visit below it, and the naive treatment pins it to the later end, which biases every duration late by construction rather than merely imprecisely. Fitted as interval-censored data — the crossing recorded as lying somewhere inside a window rather than at a point (24) — at the most sensitive plating, the tenth percentile of the first crossing falls from 2.95 to 1.82 days and the twenty-fifth from 11.24 to 9.64; across the treated arms at all four platings the twenty-fifth percentile falls from 5.77 to 3.67 days, a 36 per cent shortening. The non-parametric estimate behaves differently, and the difference is instructive. Fitted on the half-open interval the data actually give — the crossing happened after the last visit that read above the floor, and at or before the first that read below — the Turnbull curve agrees with the naive Kaplan–Meier at every visit, to floating-point precision. That is not a coincidence: the crossing intervals are the visit gaps themselves, so between visits there is nothing left for a non-parametric estimator to identify, and the naive pinning costs nothing at the visits. What the pinning costs is everything between them, which is where the parametric fit puts the shortening above. A duration read off the naive curve is therefore too long, and by about a day at the depths that matter.

What the adjustment establishes is descriptive and is worth stating as such: a continuous measure of where the cultures began accounts for the laboratory term at least as well as the laboratory label does, and one institute resists even that. What can be tested, because flasks are exchangeable across laboratories under the null, is how much of each quantity the laboratory owns. It owns 87.0 per cent of the variance in starting density (permutation $p < 0.0002$) and 36.8 per cent of the within-arm kill rate ($p = 0.0018$). It owns the duration endpoint outright, since three laboratories produce none at all.

6. The same boundaries hold in an experiment the framework never saw

Every section so far tests the framework on the deposits it was built from, and all of them are *Mycobacterium tuberculosis*. A search of five general repositories, the tuberculosis consortia, the persistence literature, food microbiology and the hollow-fibre field returned one deposit that carried all three fields the boundaries require and could be analysed cold.

That was true of the search as described. It is no longer true of the analysis repository, which now holds 40 ingested deposits. Of those, 17 permit the boundaries to be computed once the floor is established on the same tiers this paper uses for its own: stated by the depositor, derived from a recorded plated volume, or inferred from a pile-up of counts on a plate-plausible value. Across those, the condition the boundaries imply — that no series may present as a measurement a reduction deeper than its own floor allows — holds in 2 210 of 2 210 series in seven species. Those deposits are described in the data release rather than here, because none was held out. The point of this section is a deposit opened after every boundary and threshold was fixed, and only one qualifies on that ground.

Dubey and colleagues (2026) report amoxicillin-clavulanate against *Escherichia coli* in a hollow-fibre system (16, 17). Their Methods state 100 µL plated with counts per mL, so L = 10 CFU/mL is derived rather than inferred, and the file corroborates the derivation: a 100 µL plate reporting per mL can return only multiples of ten, and all 229 genuine counts in the deposit are multiples of ten, the smallest exactly ten. Sixty-nine further entries read as one, which no 100 µL plate can produce; they are the deposit's placeholder for below the floor.

Across the 20 cultures with a measured day-zero density, headroom runs from 4.85 to 5.45 log10. Endpoints at 1, 2, 3 and 4 logs are reachable for every culture. A 5-log endpoint is unreachable for 5 of 20. A 6-log endpoint is unreachable for all 20 (Table S8). On data it was not built from, the framework therefore locates the depth at which a sterilisation claim in this experiment stops being demonstrable.

Two features of that deposit are worth recording because they are the failure modes this framework is meant to catch. Its Methods give a nominal inoculum of 10^5 CFU/mL while the file measures a median of 1.215×10^6, so taking *No* from the Methods rather than from the data would have placed a full order of magnitude into every boundary. And its treated arm reads at or below the floor already at day zero while the paired control reads about 10^6, so that sample was drawn after exposure rather than before it.

7. Read the plate late enough and a 32-fold dose difference disappears

The deposit tests apramycin at 1, 4, 8, 32 and 128 μg/mL (14, 15). The 1 μg/mL flasks grow rather than die, so that arm is left out of the dose comparison. What remains is 4 to 128 μg/mL: 32-fold, five doublings. Across that range the survivor counts spread out 6.9-fold at day 3, 48.2-fold at day 7 and 5.2-fold at day 14 (Fig. 3, Table S9). The spread opens, then closes again. An experiment read at 14 days would report this drug as insensitive to a 32-fold change in dose. Each interval is fitted on its own, resampling all three replicate counts with replacement at both ends of every interval, 20 000 draws. Over days 0 to 3, each doubling of the dose adds +0.059 log10 per day to the kill rate (95 per cent percentile interval +0.033 to +0.083). Over days 3 to 7 it adds +0.033 (+0.008 to +0.059). Over days 7 to 14 it adds −0.025 (−0.035 to −0.015). Early against late is a firm difference (+0.084, +0.055 to +0.111). Early against middle is not (+0.026, −0.011 to +0.062). So concentration dependence is positive early, still positive but smaller in the middle, and reversed late — and the reversal, not just the shrinkage, is what a 14-day readout would be measuring.

We do not claim the late slope is truly negative. By day 14 the highest arm is down to 65 CFU/mL, and the deposit states no limit of quantification and no plated volume, so its floor cannot be worked out. Put the floor at 100 CFU/mL or above — the value 10 µL plating would give, an assumption and not a derivation — and that arm has already dropped below what its plate can count. Its readings are then censored: recorded only as "below the floor", true value unknown. The rate read off them is a lower bound. What holds whatever the floor turns out to be is that the separation collapses.

8. One culture, two plated volumes, two tolerance labels

Every result above reads data somebody else generated for another purpose. This one reads an experiment built to try to break the two boundaries *N*_reach and *N*_id — the starting densities that decide whether an endpoint is reachable and a label means anything — with its predictions written down before a plate was counted. *Escherichia coli* ATCC 25922, of the American Type Culture Collection — the reference strain the standard names for quality control — against ciprofloxacin at ten times the concentration that stops it growing. Three seeding densities, two plated volumes, three flasks each, 216 plate readings.

The middle arm was seeded on purpose at the 5×10^5 per mL the standard specifies. The densities actually reached were 6.4×10^4, 4.4×10^5 and 3.8×10^6 per mL. **At the standard inoculum plated at 10 µL, the deepest reduction the three flasks could report ran from 3.71 to 3.93 logs — each arm's own ceiling, not the drug's** — while those same three cultures plated at 100 µL ran from 4.88 to 5.05 (Table S10).

Every flask is paired with itself across the two platings, so the comparison is within a culture rather than between cultures.

The gain the pipette buys is exactly one log, because the floor at the pooled 100 µL plating is one tenth of the floor at the pooled 10 µL plating. The nine paired gains are not exactly one log: they run 0.87 to 1.19 (Table S10). The difference is not the drug and not the flask. It is that each plating measures its own starting density from the same culture, and the two measurements disagree — 425,000 against 565,000 per mL in one flask, 255,000 against 375,000 in another. That disagreement is this paper's own argument turned on its own experiment: N_0 is not a constant a protocol supplies, it is a measurement with error, and the headroom computed from it inherits that error. Reported per flask, the gain never falls below 0.87 $\log_{10}$, so the four-log endpoint moves from unreportable to reportable in every one of the three cultures at the standard inoculum.

That much is arithmetic, as Box 1 says. So is the next step. When both platings of one sample rest on their own floors, the two fractions written down are L_1/N_0 and L_2/N_0 — two different floors over one starting density — so they can fall either side of a class threshold and give one culture two labels. **The quantity that could have come out zero is how often a real experiment lands in the window where those two fractions straddle the cut**, and it depends on where the cut is (Table S11). Put the cut at one per cent and the window is nearly shut. Put it at one in a thousand, which is where the clinical classification analysed above cuts its lowest class, and six of 54 sample-times receive two different labels from one culture — one in nine.

That figure is computed with a single starting density per flask, and it has to be, because the sentence above says L_1/N_0 and L_2/N_0 — one denominator. The experiment can also be read the way a laboratory would actually run it, with each plating series carrying the starting density it measured for itself, and then the count is nine of 54 rather than six. The three extra are not the pipette. They are the two platings disagreeing about how much culture went in: across the nine flasks those two estimates differ by 0.74- to 1.53-fold, median 1.12. Both numbers belong in the record. Six is what the plated volume does on its own, and it is the claim this section makes; nine is what a laboratory would see, because in a real experiment the starting density is measured too, and it is measured with error.

Discussion

What fails, and where

Every clinical microbiologist knows that a negative culture taken under antibiotic exposure is not proof of sterility. That is not what this paper reports. What it reports is that the number brought in to replace that judgement carries the same defect — and once it is a number, it is harder to see.

The minimum duration for killing was built to be scale-free. It reports a fraction, and a fraction divides out the starting density. That is the property meant to make it the time-side counterpart of the minimum inhibitory concentration.

The property holds in the definition. It fails in the measurement, for a reason that is arithmetic, not statistical. A q-log reduction can only be seen where q logs are visible below where you started. When the last count rests on the plate's floor, the fraction written down collapses to L/No — the floor divided by the starting culture. The drug has dropped out of it.

Fifteen per cent of the clinical isolates examined here could not have shown the deepest endpoint whatever the drug did to them, and every one of them is recorded as having failed to show it. Eighteen isolates ended at the same censored value — recorded only as below the floor, true value unknown — and their tolerance labels are reproduced exactly by a single threshold on their starting density. To see why, sweep the true count across the range the floor allows. For twelve of them only one class fits a censored reading, so the published rule had no other label to give. For the other six the compatible range straddles a threshold, so the rule cannot choose at all. Whether those cultures reached the floor is a question about rifampicin. Which label they received once they were there is a question about arithmetic, and the starting density settles it.

It is worth saying how far this reaches. The claim is scoped to the floor and no further. In the 15-day panel, 12 of 203 usable calls have a single compatible class and 6 have more than one; the other 185 rest on a fraction that was actually measured. The kill rate still carries more of the variance in crossing time than the distance does, in every multi-laboratory arm. So what fails everywhere is not every tolerance call. It is the *interpretability* of a deep log-reduction call reported without No and L. What the inoculum settles is the subset of labels for which killing had already reached the floor.

The failure is specific and it is bounded. In this deposit neither the 90 nor the 99 per cent endpoint is compromised: every isolate has the room to show both. It is the 99.99 per cent endpoint that a substantial minority cannot reach, and that is the deep endpoint the clinical classification uses. Being that specific is what makes the problem fixable rather than fatal.

This is a derivation rather than an observation, which is what lets anyone test it on their own plates. Both boundaries follow from the definitions alone, and neither refers to the drug or the strain. Give them a floor, a set of class thresholds and a starting density, and they name the affected isolates in advance. In the clinical deposit they name them exactly, isolate by isolate — though for the twelve with a single compatible class that agreement is forced by the class rule rather than an independent test. What could have gone the other way is the boundary itself: six isolates sit at N_id exactly, the strict inequality leaves them undecidable, and the deposit records all six as medium rather than low. In a deposit the framework was not built from, the same arithmetic locates the depth at which a sterilisation claim stops being demonstrable: five of twenty cultures cannot show a five-log reduction, and none of the twenty can show six.

Everything above is read backwards, out of files made for other purposes, and a derivation that only explains the past is worth less than one that survives being aimed at the future. So N_reach and N_id were used to design an experiment against themselves, with the predictions written down before a plate was counted. At the inoculum the standard specifies, the deepest reduction the three cultures could report ran 3.71 to 3.93 logs at the 10 µL plating and 4.88 to 5.05 at 100 µL. The four-log endpoint was unreportable at one plating and comfortable at the other — same flask, same afternoon. That much is arithmetic and could not have failed. What could have failed is whether a real experiment ever lands in the window where the two platings fall on opposite sides of a class threshold, and at the cut the clinical classification uses it does so for six of 54 sample-times with the starting density held to one value per flask, and for nine when each plating carries its own — one culture, two labels, and nothing between them but the pipette and the error in measuring what went in. That is reachability working as a design constraint rather than as an explanation after the fact, and it is why the calculation in Box 1 belongs before an experiment rather than after it.

What sits beneath the floor is the one thing a plate cannot report, and it has been measured. Evangelopoulos and colleagues counted the same mouse lungs three ways — colony count, a molecular bacterial load from 16S rRNA, and a most probable number — on their way to a different question (25). Under the deepest regimens the colony count reads zero in every animal. In three arms, 18 lungs in all, the plate declared every lung sterile while the most probable number on that same tissue ran from 1 450 to 18 700 per lung. The molecular load can be dismissed as nucleic acid outliving its owner. The most probable number cannot, because it is a culture, and growth in a dilution series requires an organism that is alive and culturable. Those lungs held thousands of viable bacilli, and the plate that reported them sterile could not see them. This is the region a relapse comes from, and in the mouse model used to decide which regimens enter clinical trials it is the region a colony count is blind to.

That is also where the counts feeding regimen selection are taken. A model published as a preprint while this paper was in preparation predicts relapse in the murine model from lung CFU at 28 days together with a ribosomal RNA synthesis ratio, across nine datasets, 58 regimens and 2 239 relapse observations, reaching an area under the curve of 0.90 on external validation (26). Its own reported finding is that CFU alone performs about as well as CFU with the RS ratio, once the sterilising contribution of the individual drugs is accounted for. That is what a censored covariate looks like from the outside: where the best regimens drive the count onto the floor the count stops separating them, and a term standing for drug identity takes over the discrimination the count can no longer supply. Version 1 of that preprint reports no detection limit and no plated fraction, and we do not read this as a lapse peculiar to it — the field's own consensus guideline raises neither. It is precisely why N_reach and N_id are worth writing down. A relapse model resting on a 28-day count needs that count treated as left-censored — read as below the floor rather than as a number — and h says when the question arises instead of leaving it to be noticed.

A reader may object that this runs backwards. Dropping below the limit of detection is what a *susceptible* population does; how can it make an isolate look more tolerant? The objection is a fair one, and it is answered by separating two endpoints that the literature reports side by side.

A duration endpoint asks when a fixed depth was reached. An isolate without the room that depth requires cannot reach it under any drug effect whatever, so what the file records is not a short duration but no duration at all: the isolate is censored at the end of observation, which is the top of the scale and therefore the most tolerant value it can take. Of the 217 isolates, 33 lack the range for the 99.99 per cent endpoint the classification uses, and 33 of those 33 are recorded at that ceiling, against 162 of the 184 with ample room. The isolates that could not demonstrate the endpoint are, without exception, the ones recorded as having failed to reach it.

A class endpoint asks how far the population fell by a fixed time, and there the arithmetic does not merely preserve the direction — it inverts it. Once the final reading sits on the floor the recorded fraction is L/No, and with L fixed a *lower* starting density yields a *larger* recorded fraction. A larger surviving fraction is a higher tolerance class, not a lower one. Eighteen isolates ended at the same floor value at day 5. Their starting densities span 265-fold; their recorded apparent survival spans 265-fold. The two agree to every digit because they are the same quantity, and the labels split accordingly, twelve low and six medium.

Neither route asks anyone to mistake a sterile culture for a surviving one. In the first the assay returns no endpoint. In the second it returns the floor divided by the inoculum. What is read as biology is, in both cases, a property of the measurement.

The same arithmetic answers the objection that the starting density is fixed by protocol. It is fixed as a turbidity, and a turbidity is not a viable count. Under one written protocol the densities actually achieved differed enough to move the deepest demonstrable kill by 2.33 log10 between laboratories, and by 4.40 once the choice of plated volume is included. Those are two unlike quantities and the paper keeps them apart: 1.6 log10 of the 4.40 is exact arithmetic on $L = 1000/v$ and carries no sampling uncertainty at all, while the density part is a range over the four laboratories that deposit an untreated day-zero reading at the 100 µL plating, and its cluster bootstrap runs 0.05 to 2.33 log10. The exact part is the stronger half. The choice of plated volume is what sets the floor.

The standard is not silent on that choice, and this is the paper's sharpest finding rather than an awkwardness for it. M26-A's own section 1.3.2.5 is headed *Volume Transferred*, and it ties the volume to the endpoint with a counting rule: plate a volume such that, after the defined 99.9 per cent killing, at least ten colonies remain to be counted (10). It permits 10 to 100 µL, and gives a reason at each end — carryover above, Poisson sampling error below. Section 3.1 goes further and requires that the smallest accurately detectable count be established by serial dilution of a known inoculum. A guideline written in 1999 therefore already required both of the numbers this paper asks for: a floor established by measurement, and a plated volume chosen against the endpoint.

Applied to its own endpoint the rule bites. Ten colonies after a 99.9 per cent kill of 5×10^5 per mL needs at least 20 µL streaked; a 10 µL streak yields five and fails the rule. The requirement was there, it was quantitative, and the deposits reanalysed here do not meet it — not because their authors disregarded a standard, but because the endpoint moved out from under it. M26-A sets 99.9 per cent throughout and nowhere contemplates a four-log call; the deeper endpoints came later, and the volume rule did not travel with them.

What the tolerance phenotype tracks instead

Take the assay's contribution out and something is left, not nothing, and what is left makes biological sense. Growth state predicts the tolerance class, and it still predicts it after adjustment for starting density, which is the expected direction: isoniazid needs KatG to activate it and needs active cell-wall synthesis (27), rifampicin needs transcription (28), and a population that is not dividing offers less of what these drugs act on. The axis the classification is reading is physiological state, not drug susceptibility.

The isoniazid-resistance association behaves differently, and instructively. It is the weaker of the two on two separate counts. Unadjusted, resistant isolates carry 2.32 times the odds of a higher tolerance class. Adjusted for starting density the odds ratio is 1.31 and the interval spans one. It also does not clear its corrected threshold once the analysis is restricted to the baseline isolates, which are one per patient by construction — the only correction for repeated isolates this deposit permits, since it carries no patient identifier. The growth association survives both.

Why the association weakens is visible in the same file: resistant isolates enter the assay ten-fold lower and 26.2 per cent of them lack the room the endpoint requires. We do not know why they seed lower. The natural explanation — that resistance carries a fitness cost — is not supported here, since these isolates do not grow measurably more slowly and most carry the near-neutral *katG* S315X allele (29, 30). That gap is a finding in its own right and belongs in the next study rather than in a speculative sentence in this one. Nor is it explained away by anything else the file records: of its nine pretreatment covariates only two can legitimately be adjusted for, and neither moves the association by more than five per cent (Section 4). The two larger movements in that table both come from columns that record the exposure rather than a confounder: the mutation identity, which is missing for all but one susceptible isolate, and the isoniazid MIC, which separates the two groups completely and enlarges the seeding gap by 13 per cent instead of shrinking it.

How large the effect is

Where the inoculum is set deliberately rather than by clinical accident, the same arithmetic operates and can be measured rather than inferred. In more than a third of cross-laboratory comparisons the flask in which the population fell faster was the one that crossed below the assay floor later, and the distance-over-rate criterion accounts for three in five of them. Resampling whole laboratories rather than pairs leaves that rate poorly determined — 3.0 to 72.4 per cent — so what the deposit establishes is the mechanism and not the rate.

The variance decomposition sets the size of the effect, and it will not carry more than that. At the flask level the rate term takes the larger share in every arm examined, but the flasks are three to a laboratory and the ordering does not survive being asked at the laboratory level, so this paper does not assert it. What the decomposition does establish is that the two terms are of comparable size. A crossing time is not mostly a measurement of the inoculum, and it is not mostly a measurement of the drug either. It is a mixture in which the inoculum is a large minority — enough to reverse the ranking of two populations in more than a third of comparisons, which is the number that matters to anyone choosing between two compounds, and not enough to license the claim that the endpoint measures the starting culture.

Where we agree with the original authors, and where we differ

We agree with van Wijk and colleagues wherever the two analyses overlap. They noticed the spread in starting densities themselves. They report that the burden at the start varied between laboratories while the net effect of the drug varied less (6); that conclusion is theirs, and it anticipates part of ours. Where we part company is in what they did with readings that fell outside the quantification limits: they left them out of the numbers. That is a defensible way to report an assay. It also closes off the question this paper asks, because those are exactly the readings a duration is built from.

Vijay and colleagues report that tolerance goes with resistance status and with treatment history in the same isolates (5). We do not dispute a single measurement. Their deposit is unusually complete, and this analysis was only possible because they posted the classification, the readings behind it and the assay geometry together. What we add is this. Their deepest endpoint asks for the largest fall, so it is the one shortest of room to fall in, and how much room there is differs systematically between the very groups being compared. The link to resistance then survives neither putting starting density into the model nor cutting the data down to one isolate per patient. The link to growth survives both. So the finding is not that their classification is empty. It is that one of its two associations is carried by the geometry of the assay and by isolates that are not independent of each other.

Limitations

Four things about these deposits bound what the analysis can establish. Each of them points at the experiment that would settle the question.

The first limitation is one this paper has already acted on rather than merely declared. Flasks from one laboratory share a starting culture; repeat isolates come from a patient already counted. Every conclusion drawn from the two primary deposits was recomputed at the level where the observations really are independent, and the outcome is listed one line at a time rather than summarised (Table S12): twenty survive unchanged, six survive with much wider uncertainty, five do not survive and **have already been removed from the text above**, and one is withdrawn because the "no difference" it was tested against was already false before any data were seen. The five and the one are not claims this paper makes; they are claims it tested and dropped, and the ledger is printed so a reader can check that the dropping was done rather than promised.

The evidence between laboratories rests on six laboratories, not on hundreds of flasks. Every flask in a laboratory was seeded from the same starting culture, so for this purpose those flasks are copies of one another. The effective sample size for any comparison *between* laboratories is six, not the 67 to 85 flasks the tables list. Six is too few for the standard correction — a cluster-robust standard error — and the correction does not fail loudly when you fit it anyway. On these data it returns a *smaller* p for every institute term than the model-based p it was meant to correct. So where a comparison is between laboratories we quote the exact laboratory-level test or a bootstrap that resamples whole laboratories, and where the design cannot produce a meaningful p at all, we give no p. Three laboratories recorded crossings at the 100 μL plating and three did not; with the six split three and three, the smallest two-sided p the arithmetic can return is 0.10, whatever the biology. Some quantities do let flasks be pooled across laboratories — the variance shares, and the inversion rate under a flask bootstrap. Those keep an exact permutation test, and they are the strongest between-laboratory evidence the deposit holds.

The laboratories did not all count their starting density on the same day. Only institutes C and D deposit a day-zero reading for the treated flasks. A, B, E and F deposit day one, and by day one the ten-times-MIC arms have already lost between 0.7 and 2.4 log10. For four of the six laboratories, then, the number we are calling the exposure has some killing baked into it. We re-ran the whole comparison on a day-one exposure for every laboratory. The ordering, the area under the curve and the laboratory-level p all stayed the same, so the conclusion is robust to it. The mismatch is real all the same, and we report it rather than smooth it over.

In the six-laboratory exercise, starting density and laboratory are close to the same variable. 87.0 per cent of the variance in flask-level starting density lies between laboratories rather than within them. Taken over every flask that deposits an early density, treated and untreated arms together, the laboratory means span 3.62 log10, against a median within-laboratory standard deviation of 0.43. Adding starting density to a model that already knows which laboratory a flask came from is therefore not separating two covariates. It is closer to swapping a label for a number carrying much the same information. Telling the two apart needs a design that fixes the inoculum across laboratories.

The 15-day and 60-day panels of the clinical deposit contain the same isolates. The comparison between panels is therefore not independent, and it bounds the evidence for a difference rather than establishing it.

Starting density could sit on the causal path rather than beside it. Suppose resistance slows growth, and slow growth causes genuine tolerance. Then the thin inoculum is a consequence of the slow growth, and adjusting for it would remove real signal rather than a confounder. The asymmetry favours the confounding reading, since growth survives adjustment and resistance does not, but a design that fixes the inoculum would settle it.

What should change

Report the starting density and the assay floor with every tolerance measurement. The two together say how far down the experiment could look, and without that a log-reduction endpoint cannot be interpreted. Neither costs anything to record: one is the count that went into the flask, the other is what a single colony on the plate is worth. Four of the deposits examined in the course of this project state no limit anywhere.

This is not an aspiration. One study already does it. A murine preventive-therapy experiment writes its lower limit of detection into the deposited workbook once on every sheet, separately for each agar type, and separately for individual mice where the lower limit differed between them: 0.78 log10 CFU per lung on plain and hygromycin agar, 0.54 on thiophenecarboxylic acid hydrazide agar, with the plated volume stated beside it (31). That is better practice than anything else in this corpus. The article reporting the experiment states none of it — a full-text search returns no occurrence of "limit of detection" or of either value. So the number is not missing from the science. It is missing from the paper, and a reader who never opens the supplementary spreadsheet cannot know an endpoint was bounded. What we are asking for is one sentence in a methods section, and somebody has already done all the work that sits behind it.

The mirror image is a deposit whose modelling table carries a column named `CFU_LOD`, and that column is empty in all 272 rows (32). Somebody designed the field, declared it numeric, and never filled it in; nor does a detection limit appear in any of the eight analysis scripts deposited beside it. Put the two cases together and the diagnosis is not carelessness. The assay floor L is measured. Sometimes it is written down to a standard higher than anyone asks for. It then falls out of the record between the bench and the reader — off the end of a spreadsheet, or into a column nobody completed.

Choose the depth of the endpoint to fit the headroom actually available. The inoculum this needs can be worked out before the experiment rather than diagnosed after it. Seeding above $L \cdot \max(10^q, 1/c_1)$ makes both failure modes impossible — the endpoint that was never reachable, and the last count that lands on the floor. For this assay that is 230 000 per mL. A 99 per cent endpoint was reachable for every isolate in this deposit, and a 99.99 per cent endpoint for 85 per cent of them. Where the deep endpoint is wanted, concentrate the starting culture or lower the assay floor until the budget covers it. Where neither is possible, the shallower endpoint is the honest one. This is a design decision that is currently not being made.

Model an ordered label as ordered. A tolerance class is low, medium or high. Scoring that 0, 1, 2 and fitting a line asserts that the step from low to medium is the same size as the step from medium to high, and no one has established that. Proportional-odds regression — a fit built for ranked categories — costs nothing to run and reports in odds ratios. Where the proportionality it assumes fails, as it does here at 60 days, it says so rather than averaging two different effects into one null.

Do not densify the inoculum merely to buy headroom. Seeding higher makes a deep endpoint legal, and it also changes the experiment: a denser culture is a different physiological state, and this paper's own finding is that physiological state is what the tolerance label tracks. Where the headroom will not carry the endpoint, the shallower endpoint is the honest choice, and the growth state belongs on the report either way.

Treat an endpoint at the floor as a bound, not a value. A recorded fraction of L/N_0 is an upper limit on survival, not an estimate of it. Analysed as a measurement it manufactures differences between isolates whose final viable burdens the assay could not tell apart. It also hides the converse point: the range of true reductions a floor reading is compatible with depends on where that culture started, so the same censored reading stands for a different demonstrated kill at every starting density.

None of this requires new apparatus or a statistician, and to make that concrete the four checks are released as a small command-line tool alongside the analysis code. Give it a starting density and either an assay floor or a plated volume. It returns the headroom, states which of the 90, 99, 99.9 and 99.99 per cent endpoints that headroom can support, and reports the recorded fraction a floor-level reading would produce, labelled as the bound it is. Where a culture volume is known it also returns the viable burden a blank reading is consistent with; where it is not, it declines to compute one rather than assuming a volume. The tool exits with a failure code when a requested endpoint is unreachable, so it can be run before an experiment rather than after it.

Report physiological state as a required field. It is what the classification is reading, and it is currently the least documented variable in the assay.

Fix the inoculum, or adjust for it, before comparing tolerance across groups. The comparison that motivated this work — resistant against susceptible isolates — is confounded by a ten-fold difference in starting density, and nobody knows why that difference is there. Until it is understood, group comparisons of tolerance in clinical isolates should carry the starting density as a covariate.

Conclusion

A plate that grows nothing under antibiotic has never been evidence that the flask was sterile, and the field knows it. The remedy adopted was to stop asking whether the culture went negative and to start asking how far it fell, expressed as a fraction of where it began so that the starting culture could not matter. We find that the fraction stops being a fraction once the population reaches the floor. What the deposit records from that point is L/No, which moves with the starting density and not with the surviving population — in this deposit, for fifteen per cent of isolates at the depth the classification uses, and for the eighteen isolates whose recorded label a single threshold on their inoculum reproduces exactly, because for twelve of them no count below the floor would have changed which class the rule could return.

The consequence is not that tolerance is an artefact. It is that a tolerance call made without its headroom cannot be told apart from an inoculum measurement, and that the phenotype which does survive the correction is physiological state rather than drug susceptibility. Report the starting density and the assay floor. Match the depth of the endpoint to the range actually available. Treat a reading at the floor as a bound. Do those three things and a tolerance classification would mean the same in one laboratory as in the next. Until then, some of what the field calls tolerance cannot be separated from a record of how much culture went into the tube.

Materials and Methods

The condensed account. The full account, with every estimator and every sensitivity analysis, is Supplementary Methods in the supplemental file.

Ethics

This study reanalysed publicly available, de-identified datasets deposited by other groups. No new patient samples were collected, no new experimental data were generated, and no individual is identifiable from any material presented here. The author is an independent researcher with no institutional affiliation, so no institutional review board holds jurisdiction over the work; none was approached, and none is required for secondary analysis of de-identified data already in the public domain under open licences.

Use of generative artificial intelligence

Anthropic's Claude (Opus 5) was used under the author's direction to edit and revise the manuscript text, and to write and revise the analysis and plotting code in the accompanying repository, which draws every figure in this article from the deposited data. No figure, image or schematic was generated by an artificial intelligence tool; each is plotted by that code from values that regenerate from the public deposits. No artificial intelligence tool is an author. The author directed the analysis, verified every reported quantity against the source deposits and the regenerating pipeline, and takes full responsibility for the content of this article, including its accuracy and its originality.

Data availability

Every dataset analysed here was already public under an open licence when this work began, and none was generated for it. Each deposit is identified in Table 1 and cited in the reference list alongside the article that first described it: the clinical isolate panel of Vijay and colleagues (5, 12), the six-laboratory time-kill exercise of van Wijk and colleagues (6, 13), the evolved-clone deposit of Windels and colleagues (33, 34), the apramycin grid of Kaur and colleagues (14, 15), and the hollow-fibre Source Data of Dubey and colleagues (16, 17), which was held out of every fitting step.

The analysis code, both audit scripts, the headroom tool described above and the machine-readable receipts recording the software versions each stage ran under are deposited in a public repository (35), with the documentation needed to install and run them and a test dataset with the control parameter settings used here. Code created to generate the results and to interpret the data is included. The repository is dual-licensed, MIT for the code and CC BY 4.0 for the manuscript, figures and results, which is no more restrictive than CC BY; the deposits keep their depositors' own licences and are not relicensed here. It is at https://github.com/piranfar/comparative_model_presistance. [The commit identifier of the submitted version is to be inserted at submission.]

Every number quoted in this manuscript is regenerated from the deposits by that code, and two audit stages check it: one recomputes the pinned quantities — the load-bearing counts, every headline interval and every figure a conclusion rests on — from the tables they came from; the other reads the table legends, the figure legends, Box 1 and the Abstract that the first does not. Both report anything that disagrees, and neither pins every cell of every table, as the Reproducibility subsection states.

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The author declares no competing interests.

Vahhab Piranfar is the sole author and is responsible for conceptualisation, methodology, software, formal analysis, investigation, data curation, writing of the original draft, review and editing, and visualisation.

Figure legends

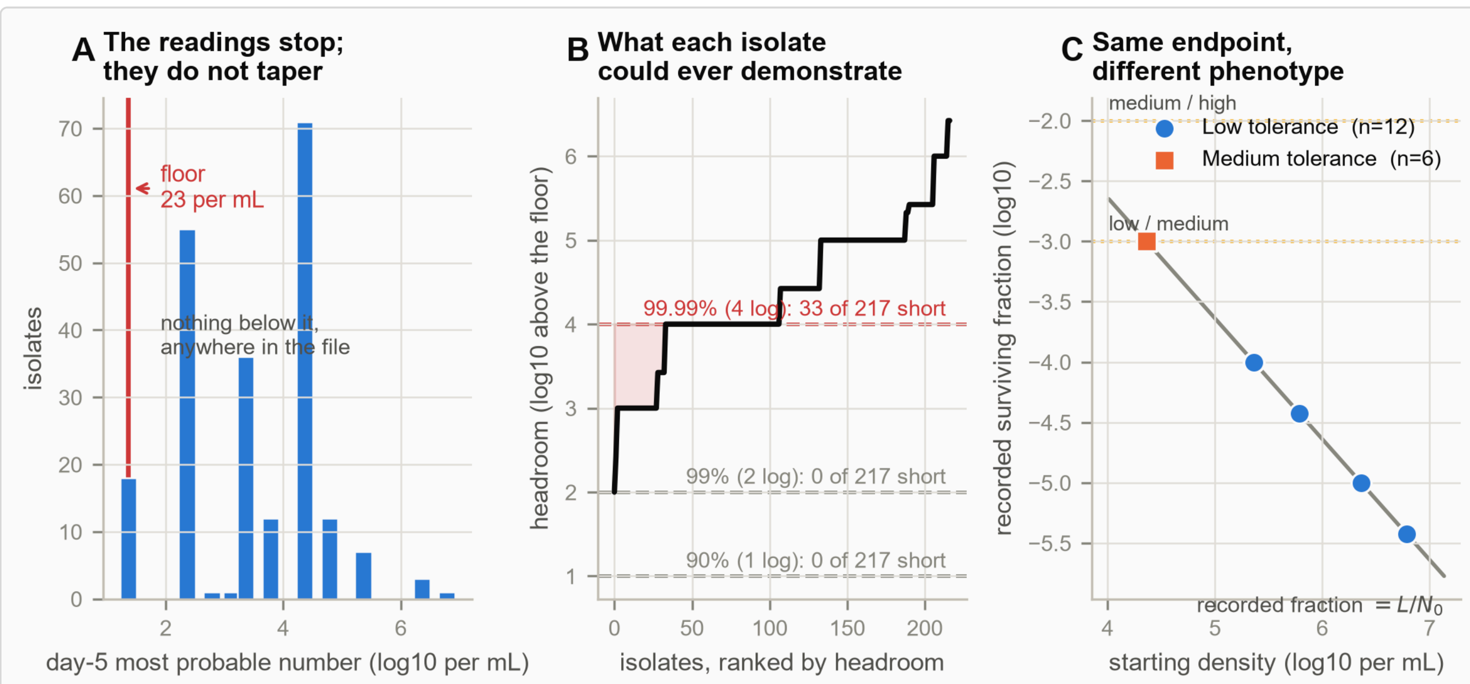

Figure 1. A log-reduction endpoint is bounded by the assay floor, and the bound decides the phenotype. 217 clinical *M. tuberculosis* isolates under rifampicin, from the deposit that carries the tolerance classification. (**A**) The distribution of day-5 most probable numbers. It stops at 23 per mL with a pile-up on it rather than tapering, and no value anywhere in the file lies below; that is the behaviour of a floor. (**B**) Headroom, the distance from each isolate's starting density down to that floor, ranked across isolates, against the depth each tolerance endpoint requires. No isolate is short of the 90 or 99 per cent endpoint. Thirty-three of 217 fall in the shaded band below the 99.99 per cent endpoint, and all 33 are recorded as failing to reach it. (**C**) The eighteen isolates of the 15-day panel whose day-5 reading was censored at the floor. Their true final counts

are unknown below L, so the assay cannot distinguish their final viable burdens; what is plotted is the recorded fraction L/No, which is what a censored reading computes, so the 265-fold span in apparent survival is exactly the 265-fold span in starting density. The dotted lines are the deposited classification thresholds. Which side of the low/medium cut an isolate falls on is settled by where it began, not by what the reading measured — though reaching the floor at all required rifampicin to cover that isolate's whole headroom, at least 3.00 logs for the shallowest and 5.42 for the deepest. For the twelve below the cut, no count the assay admits leaves any class but **low** compatible; for the six that sit exactly on it the compatible range straddles the cut and two classes remain.

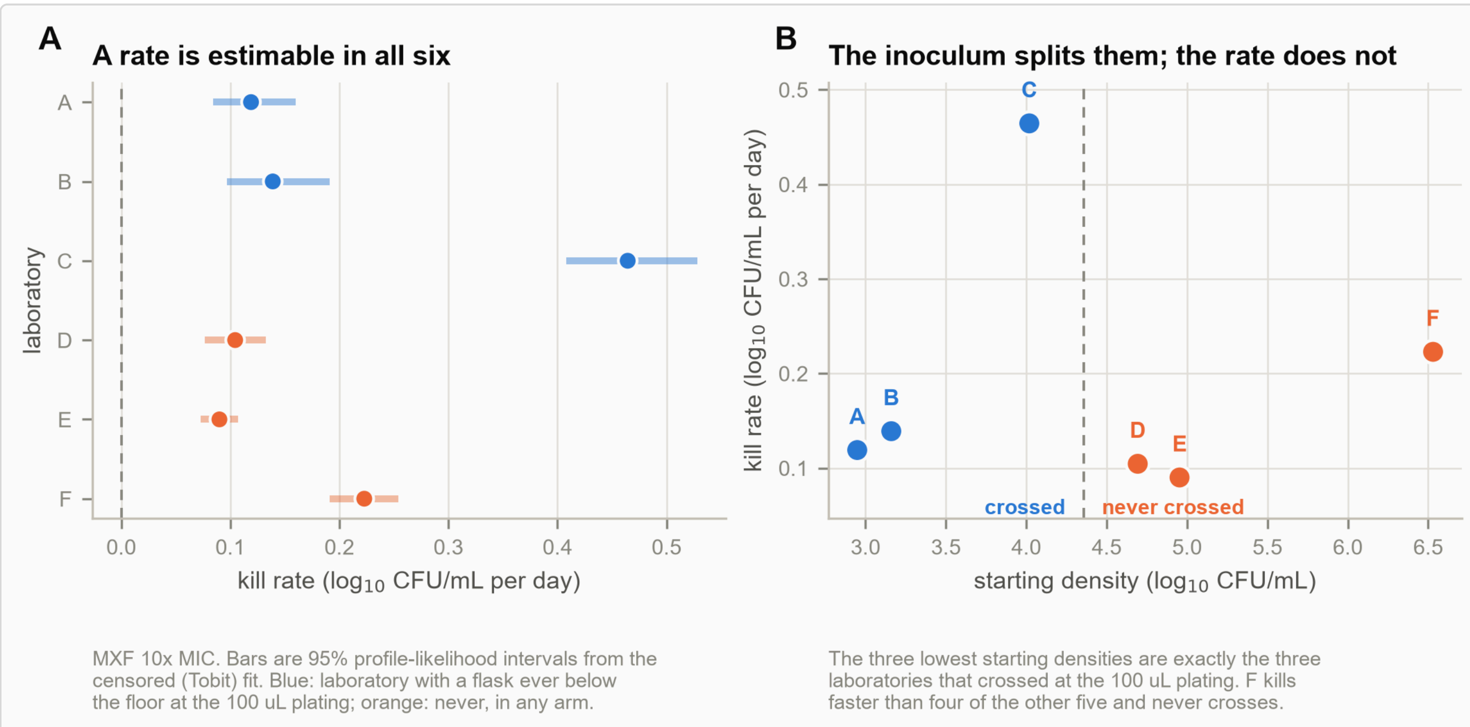

Figure 2. Where the inoculum is set by protocol, the two summaries still diverge. Six laboratories, one written protocol, one stock of *M. tuberculosis* H37Rv, moxifloxacin at ten times the minimum inhibitory concentration. (**A**) The kill rate by censored maximum likelihood with 95 per cent profile-likelihood intervals; every laboratory yields one, spanning 5.2-fold. (**B**) The two summaries against each other. The three lowest starting densities are exactly the three laboratories that recorded crossings at the 100 µL plating; the kill rate produces no such separation. The Kaplan–Meier curves for the crossing times themselves, and the movement of the laboratory terms in the descriptive Cox model, are Figure S1: both are descriptive, neither carries a between-laboratory test, and the supplement is where they belong.

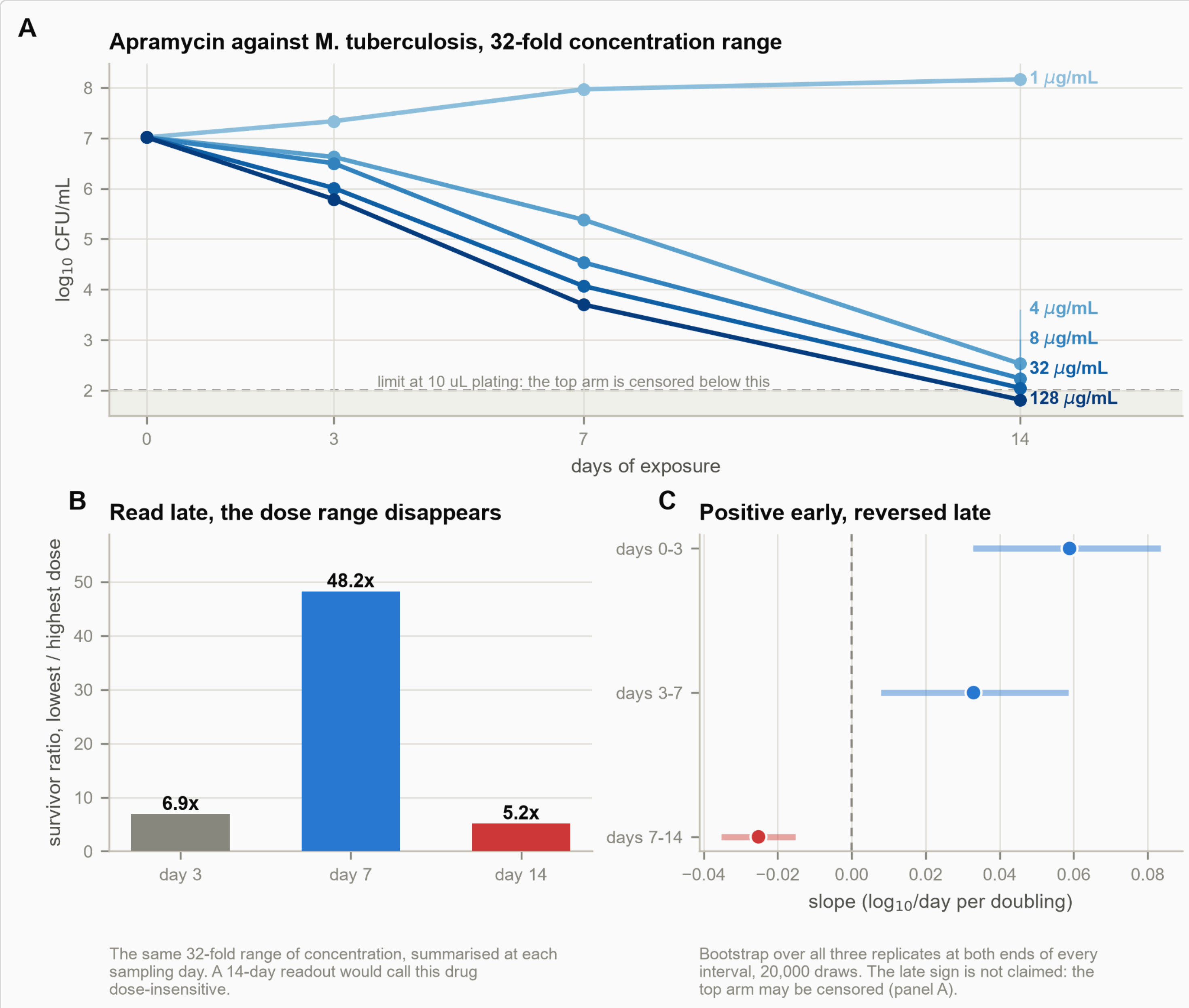

Figure 3. A late endpoint cannot resolve a 32-fold concentration range. Apramycin against *M. tuberculosis*, five concentrations, triplicate counts. (**A**) The trajectories, with an assumed floor of 100 CFU/mL — the value 10 µL plating would give — drawn as a band; this deposit records no plated volume, so the band is an assumption and is drawn to show what the assumption costs; by day 14 the highest arm lies within it, so its apparent rate over the final interval is a lower bound. (**B**) The ratio of survivors between the lowest and highest of the four compared concentrations, 4 and 128 µg/mL, at each sampling day, rising to 48.2-fold at day 7 and collapsing to 5.2-fold at day 14. (**C**) The concentration slope fitted separately in each interval from a replicate-level bootstrap, 20 000 draws.

Supplemental material

The supplemental file carries the full Materials and Methods, the analyses named above as supplementary text, and 22 supplemental tables. Table S13, Table S14, Table S15, Table S16, Table S17, Table S18, Table S19, Table S20, Table S21 and Table S22 support analyses reported there rather than in this article, and are listed so that every supplemental item is named in the manuscript.

Tables

Table 1. The five published deposits reanalysed. Four carry the analysis and the fifth is held out to test it. None was generated for this study. The held-out deposit was opened after every boundary and threshold was fixed, which the commit history of the analysis repository timestamps; the other four were selected for the fields they carry, and that selection is not separately registered.

Dataset	Organism	Drug and range	Design	Deposit	Licence
Six-laboratory exercise	*M. tuberculosis* H37Rv	moxifloxacin, isoniazid, 1x and 10x MIC	90 flasks, 2 775 readings	figshare 19766083	CC BY 4.0
Clinical isolates	*M. tuberculosis*, 217 isolates	rifampicin	6 duration endpoints per isolate	eLife 93243, suppl. file 2	CC BY 4.0
Evolved clones	*E. coli*, 126 clones	amikacin	MIC and persister fraction per clone	Zenodo 7550302 (34)	CC BY 4.0
Concentration-by-time grid	*M. tuberculosis*	apramycin 1-128 ug/mL (amikacin arm not analysed)	5 concentrations x 4 days x 3 replicates	figshare 26462791	CC BY 4.0
Held out for validation	*E. coli*, hollow fibre	amoxicillin-clavulanate	20 cultures, measured day-zero density, 100 uL plated	Nat Commun 2026 Source Data	CC BY 4.0

Table 2. The reduction each tolerance endpoint requires against the reduction the assay can resolve, in the 15-day and the 60-day prior-culture panel alike. Headroom is the distance from an isolate's starting density down to the day-5 MPN floor of 23 per mL. An isolate short of headroom cannot reach that endpoint however completely the drug worked, and every such isolate is recorded at the assay ceiling.

Prior culture	Endpoint	Isolates	Short of headroom	Fraction short	At ceiling: short vs ample
15 days	90% (1 log)	217	0	0.0%	-
15 days	99% (2 log)	217	0	0.0%	-
15 days	99.99% (4 log)	217	33	15.2%	100% vs 88%
60 days	90% (1 log)	210	0	0.0%	-
60 days	99% (2 log)	210	0	0.0%	-
60 days	99.99% (4 log)	210	7	3.3%	100% vs 95%

Table 3. Isolates whose day-5 reading was censored at the MPN floor. They share one reported floor-level observation, but their true final counts are unknown below the floor, so the assay cannot distinguish their final viable burdens; because their starting densities differ, the same reading also implies a different range of compatible fractional reductions in each. The recorded fraction is L/N0, so the spread in apparent survival equals the spread in starting density exactly, and the labels differ accordingly. The last three columns sort every call in the 15-day panel, not only the censored ones: a censored reading bounds the class from above, and sweeping the true count across the admissible range leaves twelve calls with a single compatible class and six with more than one. Reaching the floor at all is a property of the killing; which class is then compatible is a property of the starting density and the floor.

Prior culture	At the floor	Starting density spread	Recorded survival spread	Labels assigned at the floor	Measured	S
15 days	18	265x	265x	Low: 12, Medium: 6	185	
60 days	6	100x	100x	Low: 6	191	

References

1. World Health Organization. 2022. WHO consolidated guidelines on tuberculosis. Module 4: treatment — drug-susceptible tuberculosis treatment. World Health Organization, Geneva, Switzerland. https://iris.who.int/handle/10665/353829.

2. Brauner A, Fridman O, Gefen O, Balaban NQ. 2016. Distinguishing between resistance, tolerance and persistence to antibiotic treatment. Nat Rev Microbiol 14:320-330. https://doi.org/10.1038/nrmicro.2016.34.

3. Brauner A, Shoresh N, Fridman O, Balaban NQ. 2017. An experimental framework for quantifying bacterial tolerance. Biophys J 112:2664-2671. https://doi.org/10.1016/j.bpj.2017.05.014.

4. Vijay S, Vinh DN, Hai HT, Ha VTN, Dung VTM, Dinh TD, Nhung HN, Tram TTB, Aldridge BB, Hanh NT, Thu DDA, Phu NH, Thwaites GE, Thuong NTT. 2021. Ribosomal protein S1 is required for growth and antibiotic tolerance in Mycobacterium tuberculosis. Antimicrob Agents Chemother 65:e00429-21.
5. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study. Elife 13:RP93243. https://doi.org/10.7554/eLife.93243.
6. van Wijk RC, Lucía A, Sudhakar PK, Sonnenkalb L, Gaudin C, Hoffmann E, Dremierre B, Aguilar-Ayala DA, Dal Molin M, Rybniker J, de Giorgi S, Cioetto-Mazzabò L, Segafreddo G, Manganelli R, Degiacomi G, Recchia D, Pasca MR, Simonsson USH, Ramón-García S. 2023. Implementing best practices on data generation and reporting of *Mycobacterium tuberculosis* in vitro assays within the ERA4TB consortium. iScience 26:106411. https://doi.org/10.1016/j.isci.2023.106411.
7. Rabodoarivelo MS, Hoffmann E, Gaudin C, Aguilar-Ayala DA, Galizia J, Sonnenkalb L, Dal Molin M, Cioetto-Mazzabò L, Degiacomi G, Recchia D, Rybniker J, Manganelli R, Pasca MR, Ramón-García S, Lucía A. 2025. Protocol to quantify bacterial burden in time-kill assays using colony-forming units and most probable number readouts for *Mycobacterium tuberculosis*. STAR Protoc 6:103643. https://doi.org/10.1016/j.xpro.2025.103643.
8. Yamada WM, Schumitzky A, Kryshchenko A, Otalvaro J, Kim S, Louie A, Drusano GL, Neely MN. 2026. Analyzing pharmacodynamic count data that rapidly decrease to zero. CPT Pharmacometrics Syst Pharmacol 15(1). https://doi.org/10.1002/psp4.70140.
9. March VFA, Mchedlishvili K, Goig GA, Maghradze N, Avaliani T, Aspindzelashvili R, Avaliani Z, Kipiani M, Tukvadze N, Jugheli L, Bouaouina S, Doetsch A, Kalkan S, Reinhardt M, Gagneux S, Borrell S. 2025. Within-host evolution of drug tolerance in *Mycobacterium tuberculosis*. bioRxiv 2025.07.29.667394; preprint, not peer reviewed. https://doi.org/10.1101/2025.07.29.667394.
10. National Committee for Clinical Laboratory Standards. 1999. Methods for determining bactericidal activity of antimicrobial agents; approved guideline. NCCLS document M26-A. National Committee for Clinical Laboratory Standards, Wayne, PA. ISBN 1-56238-384-1.
11. Balaban NQ, Helaine S, Lewis K, Ackermann M, Aldridge B, Andersson DI, Brynildsen MP, Bumann D, Camilli A, Collins JJ, Dehio C, Fortune S, Ghigo J-M, Hardt W-D, Harms A, Heinemann M, Hung DT, Jenal U, Levin BR, Michiels J, Storz G, Tan M-W, Tenson T, Van Melderen L, Zinkernagel A. 2019. Definitions and guidelines for research on antibiotic persistence. Nat Rev Microbiol 17:441-448. https://doi.org/10.1038/s41579-019-0196-3.
12. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Supplementary file 2 to "Rifampicin

tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study" (elife-93243-supp2-v1.xlsx). eLife. https://doi.org/10.7554/eLife.93243.

13. van Wijk RC, Lucía Quintana A, Ramón-García S. 2023. *Mycobacterium tuberculosis* time kill assay data of the standardized protocol within the ERA4TB consortium. figshare. https://doi.org/10.6084/m9.figshare.19766083.v1.

14. Kaur P. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis* — raw data. figshare. https://doi.org/10.6084/m9.figshare.26462791.v1.

15. Kaur P, Ramya VK, Naveenkumar CN, Bharathkumar K, Singh M, Hobbie SN, Shandil RK, Narayanan S. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis*. Front Trop Dis 5:1413211. https://doi.org/10.3389/fitd.2024.1413211.

16. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Source Data to "Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*" (41467_2026_74323_MOESM4_ESM.xlsx). Nature Communications. https://doi.org/10.1038/s41467-026-74323-2.

17. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*. Nat Commun 17:7504. https://doi.org/10.1038/s41467-026-74323-2.

18. Cochran WG. 1950. Estimation of bacterial densities by means of the "most probable number". Biometrics 6:105. https://doi.org/10.2307/3001491.

19. Blodgett R. 2023. BAM appendix 2: most probable number from serial dilutions. Bacteriological analytical manual. US Food and Drug Administration, Silver Spring, MD. https://www.fda.gov/food/laboratory-methods-food/bam-appendix-2-most-probable-number-serial-dilutions.

20. Fisher RA. 1922. On the interpretation of χ^2 from contingency tables, and the calculation of P. J R Stat Soc 85:87-94. https://doi.org/10.2307/2340521.

21. Mann HB, Whitney DR. 1947. On a test of whether one of two random variables is stochastically larger than the other. Ann Math Stat 18:50-60. https://doi.org/10.1214/aoms/1177730491.

22. McNemar Q. 1947. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika 12:153-157. https://doi.org/10.1007/BF02295996.

23. Wilcoxon F. 1945. Individual comparisons by ranking methods. Biom Bull 1:80. https://doi.org/10.2307/3001968.

24. Turnbull BW. 1976. The empirical distribution function with arbitrarily grouped, censored and truncated data. J R Stat Soc Series B Stat Methodol 38:290-295. https://doi.org/10.1111/j.2517-6161.1976.tb01597.x.

25. Evangelopoulos D, Prosser G, Rodgers A, Dagg B, Khatri B, Bhagwat A, Gonzalo X, Kolyva A, Bertozzi G, Silva-Pereira TT, Gibbons N, Bhatt A, Sabharwal N, Perdigao J, Portugal I, Rodrigues C, Duarte R, Gomes M, Cirillo DM, McHugh TD. 2022. Data from: Comparative evaluation of viable count, molecular bacterial load and most probable number for the enumeration of Mycobacterium tuberculosis in murine tissue. University College London Research Data Repository. https://doi.org/10.5522/04/19175153.v2. Retrieved 2026-09-07.

26. van Wijk RC, Solans BP, Chaba L, Sordello S, Upton AM, Nuermberger EL, Robertson GT, Walter ND, Savic RM. 2026. Predicting tuberculosis relapse based on 28-day CFU, RS ratio, and/or drug contribution for novel regimens in the relapsing mouse model. bioRxiv 2026.07.27.740024, version 1, posted 30 July 2026; preprint, not peer reviewed. https://doi.org/10.64898/2026.07.27.740024.

27. Vilchèze C, Jacobs WR Jr. 2007. The mechanism of isoniazid killing: clarity through the scope of genetics. Annu Rev Microbiol 61:35-50. https://doi.org/10.1146/annurev.micro.61.111606.122346.

28. Campbell EA, Korzheva N, Mustaev A, Murakami K, Nair S, Goldfarb A, Darst SA. 2001. Structural mechanism for rifampicin inhibition of bacterial RNA polymerase. Cell 104:901-912. https://doi.org/10.1016/S0092-8674(01)00286-0.

29. Pym AS, Saint-Joanis B, Cole ST. 2002. Effect of *katG* mutations on the virulence of *Mycobacterium tuberculosis* and the implication for transmission in humans. Infect Immun 70:4955-4960. https://doi.org/10.1128/IAI.70.9.4955-4960.2002.

30. van Soolingen D, de Haas PEW, van Doorn HR, Kuijper E, Rinder H, Borgdorff MW. 2000. Mutations at amino acid position 315 of the *katG* gene are associated with high-level resistance to isoniazid, other drug resistance, and successful transmission of *Mycobacterium tuberculosis* in the Netherlands. J Infect Dis 182:1788-1790. https://doi.org/10.1086/317598.

31. Lai RPJ, Ammerman NC, Tasneen R, Almeida DV, Converse PJ, Nuermberger EL. 2023. Using dynamic oral dosing of rifapentine and rifabutin to simulate exposure profiles of long-acting formulations in a mouse model of tuberculosis preventive therapy. Antimicrob Agents Chemother 67:e00481-23. https://doi.org/10.1128/aac.00481-23.

32. Tabor ST, Friesen AD, Reichlen MJ, Dide-Agossou C, McGrath M, Peterson R, Ganusov VV, Robertson GT, Voskuil MI, Walter ND. 2025. Mind the gap: understanding discordance between culture- and a non-culture-based measure of bacterial burden in murine tuberculosis treatment models. bioRxiv posted 18 December 2025; preprint, not peer reviewed. https://github.com/SamuelTaborCU/Mtb-16S-rRNA-vs-CFU.

33. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Data supporting "Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence". Zenodo. https://doi.org/10.5281/zenodo.7550302.

34. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence. ISME J 18:wrae070. https://doi.org/10.1093/ismejo/wrae070.

35. Piranfar V. 2026. Analysis code, audit scripts and headroom tool for "The detection floor bounds tolerance endpoints in Mycobacterium tuberculosis".

SUPPLEMENTAL MATERIAL

Supplemental material

The detection floor bounds tolerance endpoints in Mycobacterium tuberculosis

Vahhab Piranfar

Supplementary text

Text S1. The turbidity standard fixes what goes into the flask, not what the plate can see

The inoculum for a time-kill experiment is not a free choice. You match a suspension to 0.5 McFarland and dilute it, and NCCLS M26-A (approved guideline, 1999; the body is now CLSI, which has archived the document) puts the target near 5×10^5 CFU/mL (1). If the protocol fixes the starting density, then the variable the preceding sections rest on does not vary.

Two things about the standard let counts settle this. First, a McFarland reading is a turbidity — how cloudy the tube looks, matched optically (2) — and cloudiness counts live cells, dead cells, debris and a clump of cells all as one particle (2). Converting it to CFU/mL assumes a cell size, a shape and a dispersal that *M. tuberculosis* does not oblige, and tuberculosis protocols work from a range of dilutions and target inocula (3). Second, the standard fixes what goes into the flask. What matters here is what the plate counts, and between the two sit a dilution, a transfer, and whatever clumping happens on the way.

In the six-laboratory exercise, the untreated flasks at day zero on the 100 µL plating actually started at 3.67, 4.61, 4.65 and 6.00 log10 CFU/mL, for institutes B, C, D and F. Institute E deposits no untreated day-zero reading at any volume; its series begins at day one. Institute A's three readings at that plating are flagged above the quantification limit rather than counted. Three of the four sit below the M26-A target, by 11-, 12- and 107-fold; the fourth sits two-fold above it. Every figure in this section is on that one basis — untreated, day zero, 100 µL. That is not the basis of Table S6, and the two are not expected to match.

Read as headroom at a fixed plating volume, the spread is Δh = 2.33 log10 — a 216-fold difference in how far down the assay can look, under one written protocol (Table S13). That figure is small, so its roster has to be stated. It is the largest value minus the smallest over those same four laboratories: B at h = 2.67, C at 3.61, D at 3.65 and F at 5.00. Each sits exactly one log10 below the density above it, because L = 10 CFU/mL at this plating. Institute A's other platings put it near h = 2.4, so including it would widen the spread rather than narrow it. A range over four draws is a fragile statistic: its cluster-bootstrap 95 per cent interval runs from 0.05 to 2.33 log10, and the upper limit is the observed range by construction.

The stronger version of the argument does not depend on that roster at all. Let the plated volume vary too and the spread widens to Δh = 4.40 log10. Of that, 1.6 log10 is exact arithmetic on $L = 1000/V$; it carries no sampling uncertainty whatever. The pipette moves the measurable depth further than the laboratories differ.

The obvious counter-explanation is early killing. If the drug had already acted by the day-zero reading, a spread in those readings would be kill and not inoculum. Only institutes C and D deposit a day-zero reading in both arms, and there the treated and untreated flasks agree to 0.06 and 0.13 log10, so for those two the spread is the inoculum and not early kill. The other four cannot be checked this way. Institute F shows why the check matters and why it is not decisive: its treated day-one readings average 6.62 log10, above its untreated day-zero reading of 6.00 — but the untreated culture itself grows to 6.72 over the same twenty-four hours. A day-one reading is not a day-zero reading, in either direction. A second deposit (4, 5) gives the same result against its own stated figure rather than against a standard: its Methods specify 10^5 CFU/mL and its file measures 7×10^5 to 2.8×10^6, a median of 1.215×10^6 and so 12.15-fold above nominal — computed from the unrounded median, which Table S13 displays rounded to 6.08 log10.

Distance from the McFarland reference is descriptive and is reported as such (Table S14). Sitting far below it is the intended state, since the inoculum is prepared by diluting from it, and the figure is not a measure of protocol compliance. What sets the deepest log-kill an experiment can show is the measured starting density and the real assay floor — not the turbidity the preparation began from.

Text S2. The flask that fell faster often crossed the floor later

Take two flasks in the same treatment arm from different laboratories and compare them. Across 191 such pairs — with 45 further pairs that the censoring could not settle, which are excluded rather than imputed — **66 pairs — 34.6 per cent — are inversions**: the flask in which the population fell faster crossed below the assay floor later (Table S15). Restrict to pairs whose rates differ by more than 0.10 log10 per day, so that the faster flask is unambiguously faster, and 23.0 per cent of 100 pairs still invert. The criterion *D_A*/*D_B* > *b_A*/*b_B* calls 84.3 per cent of all 191 pairs correctly, but that figure is carried by the easy majority. 65.4 per cent of pairs are not inversions, and the criterion gets 97.6 per cent of those right against 59.1 per cent of the inversions themselves. So the decomposition identifies the mechanism — a pair inverts when the distance ratio exceeds the rate ratio — without predicting which pairs invert much better than three times in five. The residual is what a single averaged slope through a two-phase kill curve cannot capture, which the Methods state as a known limitation of *b*.

The effect holds when the faster flask is required to be unambiguously faster. Raise the minimum separation between the two fitted rates and pairs whose rates barely differ drop out. At 0.02 log10 per day the inversion rate is 33.7 per cent. At 0.05 it is 29.4 per cent. At 0.10 — a difference no reader would dispute — it is 23.0 per cent of 100 pairs.

Those point estimates are firm. Their precision is not, and the difference matters. The pairs are not independent observations: 191 pairs are built from 42 flasks in six laboratories, and an interval that treats them as independent coin flips reports a precision the design does not have. Resampling whole flasks within laboratory puts the 34.6 per cent between 22.3 and 48.7 per cent. Resampling whole laboratories, which is the level at which the comparison is actually made, puts it between 3.0 and 71.3 per cent. Under the strictest rate separation the corresponding intervals are 9.4 to 42.9 per cent and 0 to 79.4 per cent. What the deposit supports is therefore that inversions are common and are explained by the decomposition. It does not support any particular rate.

The variance decomposition sets the size of the effect and it also bounds the claim. Of the variation available to move a crossing time, the distance term carries 35.2 per cent at ten times moxifloxacin, 32.3 per cent at ten times isoniazid and 17.7 per cent at one times isoniazid. The rate term carries the rest in all three. **That ordering is not established, and we do not claim it.** The flasks it is computed over are three to a laboratory and share a starting culture, so this is a between-laboratory statistic with six clusters wearing flasks as a disguise. Deleting one laboratory pushes the distance share above a half in two of the three arms — to 70.4 per cent at ten times isoniazid and 62.3 per cent at ten times moxifloxacin — and a bootstrap over whole laboratories straddles a half in all three (Table S15). What survives is the weaker and sufficient statement: the distance term is a large share of the variation available to move a crossing time, and the two terms are of comparable size.

The fourth treated arm, moxifloxacin at one times the inhibitory concentration, cannot be decomposed at all: five of six laboratories record net growth there, and only one flask in the arm returns a positive fitted rate, so there is no spread in the rate to divide (Table S15). A crossing time is therefore not mainly an inoculum measurement. It is a mixture, in which the inoculum is a large minority contribution — large enough to reverse the ranking in more than a third of comparisons, and not large enough to justify treating the endpoint as an inoculum readout in general.

Text S3. How much drug it takes and how long it takes are two different things

The framework that defines these two measures takes it as given that they are separate (Brauner et al. 2016) (6). The deposits do not prove it. They bound how far from separate the two could be, which is weaker and more honest. The 217-isolate file supports 24 comparisons of the two: how much drug it takes to stop growth against how long it takes to kill. All 24 are corrected together. Testing that many things at once throws up hits by chance, so the Benjamini–Hochberg thresholds in Table S16 — the standard correction for a family of tests — are set against 24, not against the size of any one subgroup. Four are significant before correction, where 1.2 are expected by chance, and none survives it (Fig. S2, Table S16). All four point the wrong way: a higher inhibitory concentration goes with a *shorter* killing time, which is not the direction a shared mechanism predicts. And they cluster at the deepest endpoint — the one Section 1 shows is compromised. Across all 24 tests the weakest correlation this design could have detected runs 0.14 to 0.25 depending on the subgroup; the six strongest, which Table S16 lists, span 0.14 to 0.18.

This null is narrower than it looks, and we report it as such. Take only the baseline isolates, where every isolate comes from a different patient, and the deepest 15-day endpoint gives a correlation of $\rho = -0.21$ with $p = 0.0086$. Against the family of 24, the value it has to beat is 0.0021, and it is nowhere near. Had those six baseline tests been declared a family of their own in advance, the threshold would have been 0.0083, and it would still have missed — by three and a half per cent. The family of 24 is the one this paper corrects against, so the first reading is the honest one. The second is on the record only because it points the same negative way. The axes are not shown to be independent here. They are shown not to be positively coupled, and that is what the deposits can support.

Ask the same question of 126 evolved clones sharing one ancestor (7, 8) and the answer is $\rho = +0.043$ ($p = 0.63$), against the 0.175 this sample size could have detected. Independence holds inside every nutrient group separately. Splitting by nutrient is not optional there. The concentration written on the flask is not the same exposure in every group: at 25 µg/mL, that one figure is anywhere from 0.55 to 9.47 times the inhibitory concentration that population had evolved to, and it falls on both sides of it (Table S17).

Extended results for Section 4: The label tracks how fast an isolate grows, and its link to resistance cannot be separated from the inoculum

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The label has three levels in a fixed order — low, then medium, then high — and we model it as such. Scoring the levels 0, 1 and 2 and fitting a straight line would assert that the step from low to medium is the same size as the step from medium to high, and nothing establishes that. Associations are therefore reported as odds ratios from proportional-odds ordinal logistic regression, a fit for ranked outcomes that does not assume the steps between ranks are equal. That fit makes an assumption of its own — that a single odds ratio serves both cut points — and we tested it rather than assuming it (Methods; Table S4).

These 217 isolates are not 217 independent observations. Forty-three are follow-up isolates taken during treatment from patients who also gave a baseline isolate, so up to 86 rows sit in clusters of two. But the deposit carries no patient identifier: Archive and Sample-ID are unique to each row, and none of the 48 columns repeats the way a patient key would have to. The pairing cannot be recovered, so no standard error can be clustered on the true grouping, and we do not pretend otherwise. What is available is the exact substitute, a baseline-only stratum in which every isolate is from a different patient by construction (Time_point = 0M, n = 167 at 15 days).

Why they seed lower is not established here, and the obvious explanation fails: resistant isolates do not grow significantly more slowly in this deposit (median time to OD 0.4 of 19 against 17, p = 0.24), and 85 of them carry *katG* S315X, the mutation that predominates clinically precisely because it is close to fitness-neutral — it costs the bacterium almost nothing to carry (9, 10). What the deposit can rule out is less than we first reported, and the correction runs in our favour. Seven of the nine pretreatment covariates the file carries cannot adjust this association, and they fail in two different ways. Five of them are recorded almost only for resistant isolates, so their missingness *is* the exposure: the two susceptibility calls are present for 82 of 84 resistant isolates and for none of the 119 susceptible ones, the two Mykrobe calls for 76 and none, and the mutation identity for 79 and one. Adjusting for such a column adjusts for resistance itself, which is why the two susceptibility rows agree to three significant figures — they are missing on exactly the same rows and hand the fit the same design matrix. That also disposes of the largest attenuation we previously quoted, 22 per cent for the mutation identity, which was an artefact of the same circularity. The two Mykrobe columns hold a single value wherever they are recorded in this stratum, so no adjusted coefficient exists for them at all and Table S18 prints none.

The other two are the isoniazid and rifampicin MICs, and their missingness runs the other way: they are recorded for all 119 susceptible isolates and for 67 of 84 resistant ones, so what is absent sits inside the exposed group rather than across the contrast. That does not make them confounders. The isoniazid MIC separates the two groups completely in this deposit — no resistant isolate overlaps any susceptible one — so it is the exposure measured on a graded scale rather than something to adjust the exposure for. The fit says so: the standard error on the resistance coefficient inflates from 0.103 to 0.181, and the complete case drops 17 resistant isolates and no susceptible one, leaving 186 rows against 203. It is also the second-largest movement in Table S18, and it runs the wrong way for a confounding explanation: with the isoniazid MIC in the model the seeding gap grows from −0.85 to −0.96, an amplification of 13 per cent rather than an attenuation. The rifampicin MIC, which does not separate the groups, moves it by less than half a per cent.

The remaining two are missing at rates unrelated to susceptibility and can be used: the growth proxy and months on treatment. Adjusting for either leaves the coefficient at −0.81 or −0.82 against an unadjusted −0.85, an attenuation of at most five per cent (Table S18). The seeding gap is therefore more robust than the earlier sweep suggested, not less. The file records no referring site and no processing batch, so those cannot be tested at all. The association between resistance and a low starting inoculum is real, large and unexplained, and we record it as such.

Extended results for Section 5: One written protocol does not give every laboratory the same depth to look down

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The starting densities in Table S6 need their basis stated before they are used. Each is the mean of quantified readings at day 0 or day 1 at the 100 μL plating, pooled over that laboratory's arms. Institute A deposits no day-zero reading at all, and among treated flasks only C and D do, so for four of the six the figure rests partly on readings taken after twenty-four hours of drug — by which point the ten-times-MIC arms have already lost between 0.7 and 2.4 log10. A day-zero comparison across all six is therefore not available, and the check that is available runs the other way: re-running the whole analysis on a day-one exposure for every laboratory leaves the ordering, the area under the curve and the laboratory-level exact test unchanged.

Where an interval is quoted from that fit it is a cluster bootstrap and not the model's own. For the starting-density coefficient the laboratory-clustered interval is −1.11 to −0.51, half again as wide as the model's −0.91 to −0.48 and two and a half times the flask-clustered −0.81 to −0.56; quoting the model's interval is not conservative, it is wrong in an unpredictable direction. For the laboratory contrasts the laboratory bootstrap is the wrong instrument, since it keeps each laboratory's flasks together and the contrast barely moves, so the flask-clustered interval is quoted there instead. The crossings behind each laboratory's coefficient are also worth stating plainly: 43 of 48 series in institute C against 1 of 48 in institute E.

Supplementary figures

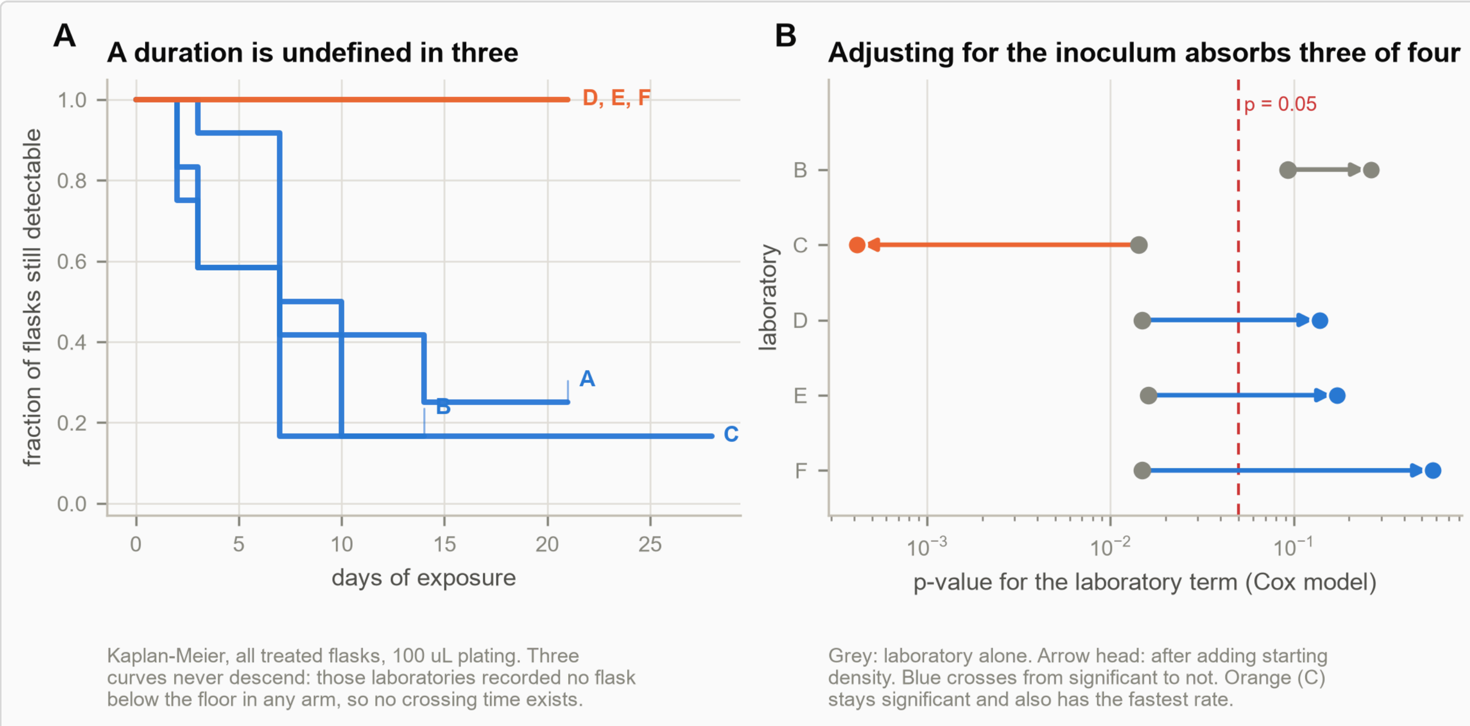

Figure S1. The descriptive survival picture behind Figure 2. The same six laboratories, one written protocol, one stock of *M. tuberculosis* H37Rv, moxifloxacin at ten times the minimum inhibitory concentration. (**A**) Kaplan–Meier curves for time to the first observed crossing below the assay floor at the 100 µL plating. Three curves never descend: those laboratories recorded no flask below the floor in any arm at that plating, so their first-crossing times are right-censored throughout. (**B**) The model-based p-value attached to each laboratory in a descriptive Cox model before (grey) and after (arrow head) starting density is added, showing how far the two predictors overlap. Institute A is the reference, so the model carries five laboratory terms; four of them move toward p = 1 on adjustment — B from 0.093 to 0.264, D from 0.015 to 0.138, E from 0.016 to 0.172 and F from 0.015 to 0.576 — while institute C moves the other way, from 0.014 to 0.00041. Those p-values are plotted because they are what moves, not because they are valid: the laboratory label is constant within its own cluster and there are six clusters, so no between-laboratory test is available and none is claimed. Blue marks a term crossing the conventional threshold, which is a description of the movement and not a finding. Both panels stood in Figure 2 in an earlier version and are here because each had to disclaim itself in its own legend. Colour, and the split between laboratories that ever crossed and those that never did, are shared with Figure 2 by construction: both figures read the assignment from one function.

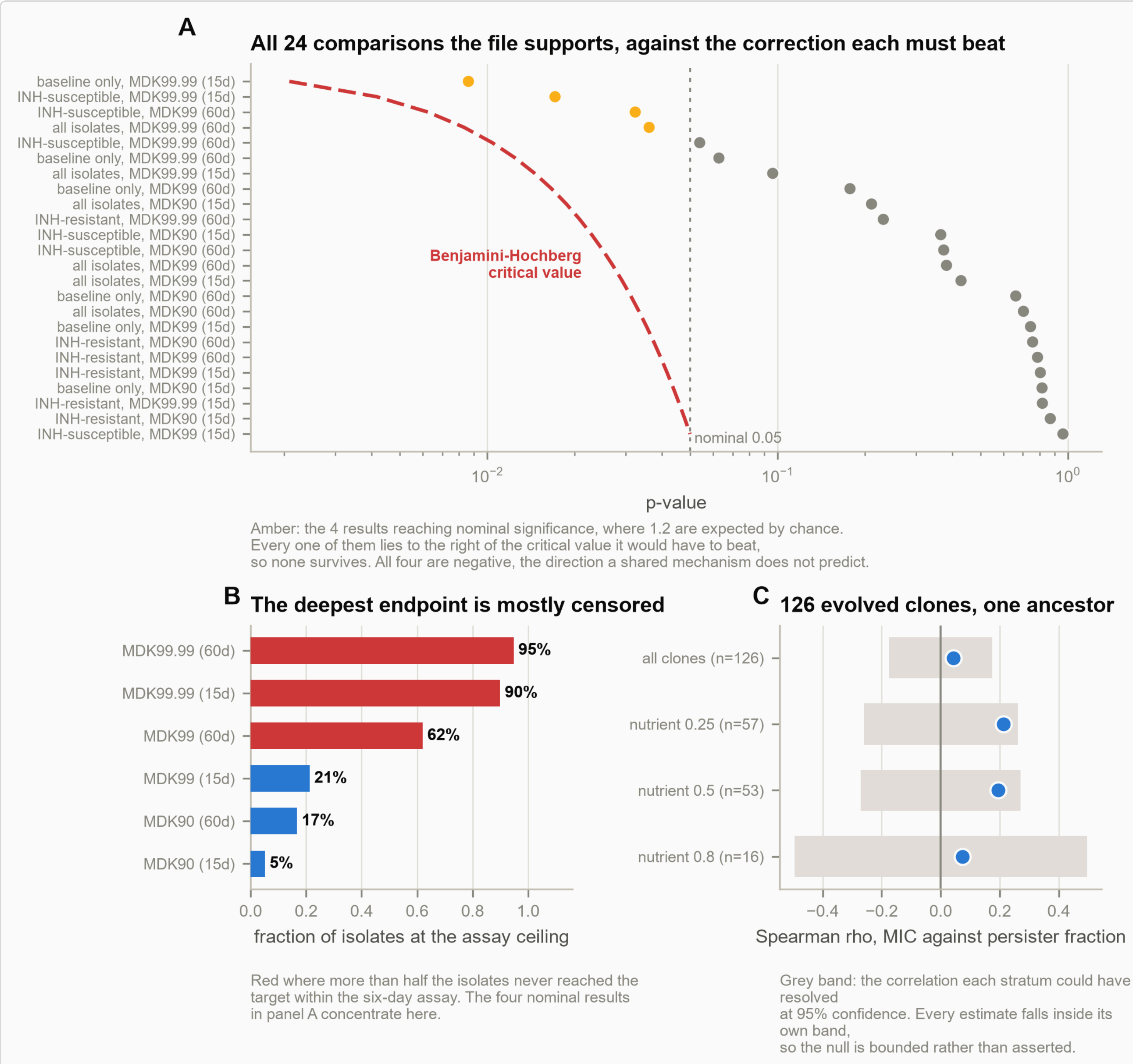

Figure S2. Resistance and tolerance occupy separate axes, in two designs. (**A**) All 24 comparisons between the concentration axis and the duration axis that the 217-isolate file supports, each against the Benjamini–Hochberg critical value it would have to beat. Amber marks the four reaching nominal significance; none survives. (**B**) The censoring behind them: the fraction of isolates at the assay ceiling rises with endpoint depth, and the nominal hits concentrate where censoring is heaviest. (**C**) The same question in 126 evolved *E. coli* clones, by nutrient stratum.

Supplementary methods

Datasets

Five published deposits are analysed (Table 1). Four carry the analysis and the fifth is held out to test it. None was generated for this study and all are openly licensed. The held-out deposit was opened only after every boundary and threshold was fixed, which the commit history of the analysis repository timestamps; the other four were chosen for the fields they carry, and that choice is not separately registered.

Clinical isolates with a deposited tolerance classification. 217 *M. tuberculosis* isolates assayed under rifampicin, each carrying a minimum inhibitory concentration, minimum durations for 90, 99 and 99.99 per cent killing at 15 and 60 days of prior culture, most probable number readings at days 0, 2 and 5, a growth-rate proxy, isoniazid susceptibility with the resistance mutation where present, months on treatment, and the authors' own tolerance level for each isolate (Vijay et al. 2024; eLife 93243 supplementary file 2) (11). The killing assay runs for six days, so an isolate not reaching its target within that window is recorded at the ceiling. Of the 217 rows, 43 are follow-up isolates from patients already represented, so a baseline-only stratum is analysed separately.

The six-laboratory exercise. *M. tuberculosis* H37Rv from one stock, distributed with one written protocol to six laboratories blinded and labelled A to F (van Wijk et al. 2023; figshare 19766083, CC BY 4.0) (12, 13). Moxifloxacin and isoniazid at one and ten times the minimum inhibitory concentration, an untreated control, three flasks per arm, sampled to day 21 or 28. Each sample was plated at four volumes: 100 µL in quadruplicate, 10 µL as four drops, 10 µL as a single drop, and 2.5 µL as a single drop. In the deposited file the colony count is already expressed per millilitre, confirmed by the smallest count recorded at each volume being exactly 1000 divided by that volume, which is one colony per millilitre. The minimum reportable positive count is therefore a property of the plated volume — one colony in that volume — and equals 1.0, 2.0, 2.0 and 2.6 log10 CFU/mL respectively, so four platings give three distinct floors spanning 1.6 log10 within a single flask-visit (14). The identity is exact for 1 055 of the 1 068 readings at 10 µL; the thirteen exceptions all come from one laboratory and do not lie on the lattice its recorded volume implies. The file holds 2 775 readings, of which 17.9 per cent are flagged below the floor and 24 above it; the analysis set, after dropping above-limit flags, unusable values and pre-treatment visits, is 2 580 readings, of which 19.3 per cent are flagged.

Evolved clones. 126 *Escherichia coli* clones from a parallel evolution experiment under amikacin, each carrying an endpoint minimum inhibitory concentration and a persister fraction measured on the same clone, labelled by the antibiotic concentration and nutrient level its population evolved under (Windels et al. 2024; Zenodo 10.5281/zenodo.7550302, CC BY 4.0) (7, 8). Six of its 595 surviving fractions are written as exact zeros, marking a below-limit reading without naming a limit; they fix no floor, so this deposit is one of the two for which observability labels are refused.

A deposit held out for validation. Amoxicillin-clavulanate against *Escherichia coli* in a hollow-fibre infection model, with 100 μL plated and counts per mL so the assay floor is derived rather than inferred, measured day-zero densities for every culture rather than a nominal inoculum, and below-limit readings retained as a placeholder (Dubey et al. 2026; article Source Data, CC BY 4.0) (4, 5). It was located by a search of five general repositories, the tuberculosis consortia, the persistence literature, food microbiology and the hollow-fibre field, and was the only deposit found carrying all three fields the boundaries require. No boundary, threshold or modelling choice in this paper was informed by it: it was opened after all of them were fixed, and Section 6 is the only place it tests anything. What cannot be documented is the search itself. The repositories were searched interactively and no query log, access date or screening count was kept, so this paper reports the outcome of that search — one deposit carrying a measured per-culture starting density, a recorded plated volume and per-culture time-course counts together — without being able to evidence its extent. The claim that no deposit was selected after its result was known rests on the commit history of this repository rather than on a registration, and readers should weigh it accordingly. It does appear descriptively — a row of its own in Tables 2, 5, 9, 15, S5 and S7, and a table of its own in Table S8 — which is reporting rather than fitting.

Concentration-by-time grid. Apramycin and amikacin against *M. tuberculosis* at 128, 32, 8, 4 and 1 μg/mL, triplicate log10 CFU at days 0, 3, 7 and 14, with a concurrent drug-free control at every visit (Kaur et al. 2024; figshare 26462791, CC BY 4.0) (15, 16). The amikacin arms are not analysed here. Within apramycin the 1 μg/mL arm shows net growth rather than killing and is excluded from the dose comparison, leaving 4 to 128 μg/mL as the range compared (Section 7).

Two kinds of limit, kept apart

No deposit analysed here reports a validated limit of quantification with a value. The six-laboratory file names the concept in the definitions of its BQL and AQL columns but defines it as whether the plate was countable, which is an operator's judgement rather than a validated limit. Throughout this paper L is therefore an **operational assay floor**: where a plated volume is recorded it is the **minimum reportable positive count**, one colony in that volume, and where none is recorded it is inferred from the deposit and is labelled as inferred. The term limit of quantification is used only in reporting what a source states.

Those two names are the precise ones and are used where L is defined or its provenance discussed. In running text L is the **assay floor**, and that is the only short form used, so that "boundary" is left free for the two derived boundaries of the next subsection and never denotes a value. The event a time-to-event analysis records is correspondingly a first observed crossing below the assay floor. What L is in each deposit, how it was arrived at, and how replicate plates and plated volumes were treated is tabulated deposit by deposit (Table S19).

Two distinct censoring problems arise and are handled separately, because conflating them is the error this paper is about.

A **count below the assay floor** is left-censored: the population is somewhere below a known value. In the six-laboratory deposit that value is a property of the plated volume, as above. In the clinical deposit the readings are most probable numbers taking the discrete values of an MPN table (17, 18), and the day-5 column bottoms out at 23 per mL with a visible pile-up there; that value is treated as the floor, and the first Results section reports the evidence for it.

That deposit states no limit of quantification, so the floor is inferred rather than read off, and the inference is bounded rather than asserted. Three things support it: 23 per mL is the smallest value anywhere in the file, nothing lies below it in 1 932 readings, and the minimum shifts by exactly a factor of ten per visit — 2 300, 230, 23 — in all three culture ages, which is the signature of one dilution series applied to a differently diluted sample at each timepoint.

The floor is therefore visit-specific, and it does not follow that every visit has one. A floor is a value readings stop at, so it shows as a pile-up on the lowest rung with nothing below; the ten-fold shift alone only shows the dilution changing. Applying the rule of the next subsection visit by visit, the day-5 column has that signature in every panel and the day-2 column has it at 15 days, but the day-0 column does not: its minimum of 2 300 per mL occurs exactly once in each panel, with the next value at 6 100, which is the tail of a distribution rather than a floor. The verdict for day 0 is that no floor is evidenced. The starting density is therefore an observed value throughout, and no quantity in this paper that conditions on *No* needs a censored specification. Since none of that is proof, the load-bearing count is recomputed at every value a three-tube MPN table returns below 23. The number of isolates lacking four logs of headroom is 33 at a floor of 23 and does not fall below 28 at a floor of 3.0, the lowest such table returns. The conclusion holds across the whole range, so it is not an artefact of where we place the floor, and the same sweep is run for every deposit that carries a floor (Table S20).

The six-laboratory deposit also carries the evidence for judging each reading against its own volume rather than pooling. Each sample is plated at four volumes at the same visit, so a below-limit flag can be checked against the other platings of the same flask. Of its 498 flags, 83, or 16.7 per cent, are contradicted: another plating of the same sample at the same visit, with a limit at least as sensitive, returns a quantified count above the flagged reading's limit. Those flags record a drop that missed rather than a culture that fell. Pooling the four volumes to a single limit would treat them as evidence about the population; judging each against the floor its own volume implies does not, and that is the reason the analysis is built the way it is. The cost of pooling can be priced. Pooling at the least sensitive volume would discard 126 genuine quantified counts, 5.9 per cent of them, and pooling changes the contradicted-flag rate itself: 16.7 per cent judged per volume, against 43.2 per cent pooled at 10 CFU/mL and 15.3 per cent pooled at 400. The audit's own answer depends on the choice, which is the strongest reason to make it explicitly.

A **crossing time not observed within the study** is right-censored: the event has not happened yet, which is not the same as the event having no time. A flask that never fell below the floor contributes the information that its crossing time exceeds its last visit, and enters the survival likelihood as such. It is never recorded as missing.

Floor posterior, refusal, and what is not learned

The algebra of headroom and observability is closed form once L is known. What requires inference is L itself, and the two cases are not alike. A volume-derived floor — one colony in V µL is $1000/V$ per mL, and the observed minimum equals that value — is a point mass, with nothing to infer. A floor inferred from a pile-up on a most-probable-number rung carries a discrete posterior over the rungs at or below the observed minimum; the interval quoted is the 95 per cent highest-posterior-density support.

That distinction is turned into a rule rather than a caveat. Where the verdict is that no floor is evidenced, or where the support spans more than one log10, the observability labels are **refused** rather than reported (Table S1). Surviving fractions written as exact zeros with no named L fall in that refused class, as does a deposit whose minimum occurs once in a continuous tail. The clinical deposit passes the rule with a support of 0.40 log10; two of the five deposits fail it and are labelled accordingly. This paper fits nothing to predict an MDK: once L is in hand the rest is derived.

Causal mediation of the resistance association

The attenuation of the resistance coefficient once starting density enters the model is reported in the Results as a descriptive ratio, which is what it is. Separately, a linear product-of-coefficients mediation decomposes the total effect of isoniazid resistance on the day-5 tolerance class into an average causal mediation effect through log10 N_0 and an average direct effect (19, 20), with bootstrap percentile intervals from 5 000 resamples, repeated on the baseline stratum and on two binary collapses of the outcome.

The refit that answers the objection about the mediator being the outcome's own denominator is reported in Section 4. Its stratum is defined here: an isolate is excluded from it when its day-5 reading is at or below the floor of 23 MPN per mL, which is the condition under which the recorded fraction is L/N_0 and the class carries no measured numerator. Eighteen of the 203 meet it. The refit is the same estimator on the remaining 185, with the same seed and the same 5 000 resamples, so the two are comparable term by term.

Sequential ignorability is assumed and is not testable here, so the estimate is accompanied by the sensitivity that matters: the residual correlation between mediator and outcome errors at which the point estimate crosses zero, which is about −0.23. The outcome is an ordinal class scored linearly, as in the association family; the binary collapses are reported because they do not depend on that scoring. No claim is made that unmeasured confounding is absent. The estimate replaces the attenuation percentage as the quantity cited for the pathway (Table S5).

Fold-change figures such as 216-fold and 265-fold are ten raised to the unrounded log10 difference before display rounding, so that Δh = 2.3349… is reported as 216-fold rather than as 10^2.33.

What each analysis was fitted to

A dozen different denominators appear in this paper and each is correct for its own analysis, so every one of them is derived in a single place (Table S2). The reductions are few and they are named: the MDR tolerance label is a fourth, unordered category and is dropped wherever an ordered outcome is fitted, taking 14 rows from each panel; six 60-day labels are absent before that, so the 60-day panel loses 20 in all; the growth proxy is missing for one isolate; and five of the 72 treated flasks in the six-laboratory deposit carry no usable starting density, which is why a comparison over 72 flasks is reported on 67. Of the 30 laboratory-by-arm cells, 29 retain enough quantified readings to fit a rate.

The series counts in the six-laboratory deposit differ for the same reason. Each of the 90 flasks is plated four ways, giving 360 series of which 288 are treated; the descriptive Cox drops a further 27, 26 for carrying no quantified reading at day 0 or day 1 in their own plating and one for reading below its floor at day zero, leaving 261.

Whether these exclusions are plausibly ignorable is tested rather than asserted: dropped rows are compared with retained rows on starting density and susceptibility (Table S21). The MDR exclusion differs in susceptibility by construction. The only difference not by construction is in the 60-day panel, where the 20 dropped rows sit higher in starting density.

Dynamic range, and the two boundaries it sets

For a culture starting at N_0 against an assay floor L, two quantities are used and are kept apart, because they are not interchangeable:

$H = N_0/L$, a ratio; $h = \log_{10}(N_0/L)$, the **headroom**, in log10 units.

The headroom is the deepest reduction the assay can resolve. It is fixed by the dilution scheme and the starting culture before any drug acts, and variation between cultures is reported on h rather than on N_0, as $\Delta h = \max(h) - \min(h)$ with the corresponding fold difference $10^{\Delta h}$. That distinction is not notational here. Where L is set by the plated volume, one culture read on a 2.5 µL drop and on a 100 µL quadruplicate differs by 1.6 log10 in headroom, and a figure computed from N_0 alone would discard it.

Two boundaries follow from the definitions alone.

Reachability. The smallest value the assay reports as a measurement is L, so the largest reduction expressible as a measured value is h. A q-log endpoint is therefore observable if and only if

$$N_0 \geq N_{reach} = L \cdot 10^q.$$

Identifiability. A censored reading places the true count in $[0, L]$, so the recorded surviving fraction is at most L/N_0 — an upper bound, not a measurement. Given class thresholds $0 < c_1 < c_2$ ..., the class is determined by the data only if the whole interval $[0, L/N_0]$ lies inside one class. Since that interval contains zero, the only candidate is the lowest class, so the class is determined if and only if

$N_0 > N\_id = L/c_1$.

Above that boundary the lowest class is the only one compatible with a censored reading, whatever the true count was, so the class the rule assigns is fixed by N_0 rather than by the surviving population. The boundary says nothing about whether a culture reaches the floor, which depends on the drug; it says what the classification can mean once one has. Both boundaries are properties of the method and involve neither the drug nor the strain, and their ratio is $N\_reach/N\_id = c_1 \cdot 10^q$.

Seeding a culture above $L \cdot \max(10^q, 1/c_1)$ makes both failure modes impossible, which is a design criterion computable before an experiment rather than a diagnosis after it.

Estimation

A **Tobit model** fits log10 CFU/mL linearly in time by maximum likelihood (21). An observed reading contributes the usual Gaussian density; a censored reading contributes $\log \Phi((\text{limit} - \mu)/\sigma)$, the probability that it fell below its own limit. This is Beal's M3 (Beal 2001) written for this assay (22). Intervals come from the profile likelihood.

Multiple imputation draws each censored reading from the fitted normal truncated at its own limit, refits by ordinary least squares, and pools 50 fits by Rubin's rules (23). The imputation is proper: the line and the error scale are redrawn from the Tobit fit's own asymptotic distribution at every imputation rather than held at the point estimate, so the between-imputation variance carries the uncertainty in the model that generated the draws. What the agreement between the two estimators establishes is therefore that the answer does not depend on which arithmetic recovers it. It does not establish that the sub-floor distribution is normal: both make that assumption, and no reading below the floor exists to test it against. Where the assumption itself is in question, the quantity to look at is the headroom, not the fit.

The growth proxy is the deposit's time to an optical density of 0.4, in days; the deposit records no wavelength, so none is given here. Because it is a time, it runs inversely to growth rate: a larger value is a slower-growing isolate.

The tolerance label is modelled as an ordered categorical outcome by proportional-odds ordinal logistic regression (24), with starting density as a prespecified covariate and every association reported unadjusted and adjusted for it. The MDR category of the deposited label is a fourth, unordered category and is excluded; all 14 rows carrying it fall outside the susceptible-versus-resistant contrast in any case, so the ordinal and linear models are fitted to identical rows. Proportional odds is tested by a Brant test per predictor (25) and by a likelihood-ratio test against a generalised ordered logit (26) with the coefficient released; where it fails, the released partial-proportional-odds fit is reported and checked against a multinomial fit that assumes no ordering. Rows are dropped only for a missing outcome, predictor or starting density, and the dropped rows are compared with the retained ones on starting density and susceptibility. Because the deposit carries no patient identifier, the repeated-isolate structure is handled by a baseline-only stratum rather than by a random effect, and the linear models of the earlier analysis are retained as a sensitivity comparison (Table S22).

Time-to-event analysis treats the **first observed crossing below the assay floor** as the event, with series never falling below their own floor right-censored at their last visit. The event is named that way throughout and is not read as clearance, sterilisation or the end of a culture: of the series that cross, 60 per cent read above the floor again at a later visit.

The crossing time is interval-censored, not observed: it lies between the last visit above the floor and the first visit below it, and the visit schedule is coarse. Interval-censored fits (27) are therefore reported alongside the naive treatment that pins the event to the visit at which the blank plate was noticed. Kaplan–Meier (28) and Cox proportional hazards (29) are retained as **descriptive** summaries of first crossing only; their intervals are replaced by cluster bootstraps over laboratories and over flasks (30), because those 261 series come from 72 flasks in 6 laboratories and the partial likelihood treats them as independent. No p-value is quoted for a term that is constant within laboratory, since six clusters cannot support one. Proportionality is tested on Schoenfeld residuals (31, 32).

A cell is fitted only when it retains at least six quantified readings at three distinct times; below that the slope is determined by the censoring pattern rather than by the counts.

The replicate-level bootstrap (33) behind the interval slopes resamples the three replicate counts with replacement at both ends of each interval, takes their mean, refits the concentration slope on each draw, and takes percentile intervals from 20 000 draws. The resampling unit is the replicate count, not the interval, so a draw can repeat a replicate at one end and not the other. It is a draw of the three-replicate MEAN, because the mean is what the slope is fitted through; drawing a single replicate instead would attach the variance of one count to an estimate built from three. This bootstrap, and not the least-squares fit through the four concentration means that the deposited table reports beside it, is the interval quoted in Section 7 and drawn in Figure 3C. The least-squares fit is kept in the table for comparison and is labelled there: it has two residual degrees of freedom and discards the replicate scatter entirely. An interval on a proportion — an inversion rate, or a share of flasks — is the Jeffreys interval (34), which is the equal-tailed posterior under the Jeffreys prior and does not collapse to zero width when the count is 0 or n, as the normal approximation does at the sample sizes here.

What is standard and what is not. The Tobit fit, Kaplan–Meier, Cox and the interval-censored fits come from the packages named below. Four procedures do not: the Brant test of proportional odds, the partial-proportional-odds fit with a per-cut coefficient mask, the product-of-coefficients mediation with bootstrap percentile intervals, and the residual-correlation sensitivity for sequential ignorability. Those are implemented in the released analysis code rather than taken from a package — in `exp30_ordinal_tolerance.py` and `exp34_mediation.py` — and the partial-proportional-odds likelihood is validated against the proportional-odds fit it nests, agreeing to 1.0×10^{-10} in log-likelihood when nothing is released. The multinomial check uses statsmodels.

The decomposition of a crossing time

Over an interval in which the decline is close to log-linear, log10 $N(t) = a - bt$, and with ℓ the log10 assay floor and $D = a - \ell$ the distance the population starts above it, the crossing time is $T = D/b$. For two flasks,

```
log(T_A / T_B) = log(D_A / D_B) - log(b_A / b_B)
```

which is exact on this model and splits a difference in crossing time into a distance term and a rate term. An **inversion** is a pair in which A fell faster yet crossed later, which occurs exactly when $D\_A/D\_B > b\_A/b\_B$. A single slope is not a full description of a biphasic trajectory, so b is the average decline over the window fitted, and the variance shares reported below are shares of that averaged slope; averaging across decline and regrowth moves the rate term up rather than down, so the distance share is a lower bound; D is taken from the measured day 0 to 1 density rather than from the fitted intercept, because a single line through a biphasic curve extrapolates back to an intercept well below the culture the flask started from.

Multiplicity

Where a question admits more than one test, every test the deposit supports is run, and the family is corrected by the Benjamini–Hochberg procedure (35) at a false discovery rate of 5 per cent. For each test we report the correlation the sample size could have resolved at 95 per cent confidence, so that a null is bounded rather than asserted.

Reproducibility

Every number in this paper is regenerated by a script that writes a machine- readable receipt of what it computed, and a separate audit script recomputes 174 of the quantities quoted in the text from the tables they came from and reports any that disagree. That audit is not exhaustive: it pins the load-bearing counts, every headline interval and every figure a conclusion rests on, and it does not pin every cell of every table.

A second audit reads what the first cannot. The value audit parses table bodies only, so it never sees a table legend, a figure legend, Box 1 or the Abstract — and it has no way to notice that a number called a hazard ratio in one place is called a p-value in another, which is a mistake this manuscript made and a reviewer caught. The second audit checks that every number in those four regions, conventions such as the endpoint constants and the confidence level aside, occurs in a results file, a receipt or a table body; that no numeric literal carries two incompatible statistical labels and that no most-probable number is printed as a colony count; that percentages, fold-changes, corrected thresholds, spans and interval bounds follow from their own stated inputs; and that no conclusion the self-audit marks unsupported is still asserted anywhere, that every sample size traces to a line in the flow account, that every cross-reference resolves, and that no block of prose states the same thing twice, which is what a paragraph looks like when the passage its rewrite replaced was never deleted. It fails the pipeline on the findings it grades as contradictions and reports anything weaker without failing; an untraceable number in a figure legend, in Box 1 or in the Abstract is weaker, and so is a legend count its own table contradicts where the results carry that value elsewhere. Every table is generated from the results files and the manuscript is assembled from that generated material, never edited downstream of it, so a table cannot drift from the analysis behind it.

What the second audit does not cover is worth stating, because a guarantee whose edges are unstated will be read as covering everything. It is an existence check before it is a semantic one: a number that is correct somewhere in the corpus but attached to the wrong quantity passes unless some other rule reaches it. Figure legends, Box 1 and the Abstract have no staleness net comparable to the one that diffs each spliced table against the block that generated it, so a stale number there is caught only if it exists nowhere else. An error introduced into a legend inside the table-generating code is invisible to a check that compares the paper against that code's own output. And the self-audit table is the premise of the withdrawn-claim check rather than its subject: the check sees only the conclusions that table lists, so a reading retracted in the prose alone lies outside it, and a verdict silently changed from unsupported to supported would not be detected.

Analyses used Python 3.14 with numpy 2.5.0 (36), scipy 1.18.0 (37), pandas 3.0.3 (38), statsmodels 0.15.0 (39) and lifelines 0.30.3 (40).

Supplementary tables

Table S1. What is actually known about the assay floor in each deposit, and the rule that follows from it. A floor derived from a recorded plated volume is a point mass: one colony in that volume, no inference required. A floor inferred from a pile-up on a most-probable-number rung carries a posterior over the rungs at or below the observed minimum, and the 95 per cent support is quoted. Where no floor is evidenced at all, or where the support spans more than one log10, the observability labels of Section 3 are refused rather than reported -- a label is only as good as the floor it is computed against.

Deposit	Verdict	Floor used	95% support	Span (log10)	Refuse observability labels
Vijay 2024, clinical isolates (MPN per mL)	INFERRED	23	[9.2, 23.0]	0.398	no
ERA4TB, 100 uL plated (CFU per mL)	DERIVED	10	point mass	0.000	no
ERA4TB, 10 uL plated (CFU per mL)	DERIVED	100	point mass	0.000	no
ERA4TB, 2.5 uL plated (CFU per mL)	DERIVED	400	point mass	0.000	no
Kaur 2024, apramycin grid (log10 CFU per mL)	NONE	-	-	-	yes
Windels 2024, evolved clones (surviving fraction)	NONE	-	-	-	yes
Dubey 2026, hollow fibre (CFU per mL)	DERIVED	10	point mass	0.000	no

Table S2. Every analysis set in this paper, and the exclusion that produced it. The manuscript quotes a dozen different denominators, each correct for its own analysis; this is where a reader checks which is which. The MDR tolerance label is a fourth, unordered category and is dropped wherever an ordered outcome is fitted; the growth proxy is missing for one isolate; and five treated flasks in the six-laboratory deposit carry no usable starting density, which is why a comparison over 72 flasks is reported on 67. Whether these exclusions are plausibly ignorable is tested in Table S21. The last rows of each block cover sets counted in units other than isolates or flasks -- pairs, flags, cells, readings -- because a reader meeting one of those figures in the text needs somewhere to look it up too.

Deposit	Panel	Stratum	Stage	n	Excluded here
Vijay clinical	15-day	all isolates	rows in the deposit	217	-
Vijay clinical	15-day	all isolates	tolerance label present	217	-
Vijay clinical	15-day	all isolates	label is Low, Medium or High (drops 'MDR')	203	-14
Vijay clinical	15-day	all isolates	starting and day-5 readings present	203	-
Vijay clinical	15-day	all isolates	susceptibility is IS or IR	203	-
Vijay clinical	15-day	all isolates	growth proxy present (association family)	202	-1
Vijay clinical	15-day	baseline only (0M)	rows in the deposit	174	-
Vijay clinical	15-day	baseline only (0M)	tolerance label present	174	-
Vijay clinical	15-day	baseline only (0M)	label is Low, Medium or High (drops 'MDR')	168	-6
Vijay clinical	15-day	baseline only (0M)	starting and day-5 readings present	168	-
Vijay clinical	15-day	baseline only (0M)	susceptibility is IS or IR	168	-
Vijay clinical	15-day	baseline only (0M)	growth proxy present (association family)	167	-1
Vijay clinical	60-day	all isolates	rows in the deposit	217	-
Vijay clinical	60-day	all isolates	tolerance label present	211	-6
Vijay clinical	60-day	all isolates	label is Low, Medium or High (drops 'MDR')	197	-14
Vijay clinical	60-day	all isolates	starting and day-5 readings present	197	-
Vijay clinical	60-day	all isolates	susceptibility is IS or IR	197	-
Vijay clinical	60-day	all isolates	growth proxy present (association family)	196	-1

Deposit	Panel	Stratum	Stage	n	Excluded here
Vijay clinical	60-day	baseline only (0M)	rows in the deposit	174	-
Vijay clinical	60-day	baseline only (0M)	tolerance label present	168	-6
Vijay clinical	60-day	baseline only (0M)	label is Low, Medium or High (drops 'MDR')	162	-6
Vijay clinical	60-day	baseline only (0M)	starting and day-5 readings present	162	-
Vijay clinical	60-day	baseline only (0M)	susceptibility is IS or IR	162	-
Vijay clinical	60-day	baseline only (0M)	growth proxy present (association family)	161	-1
Vijay clinical	15-day	other sets quoted	isolates with a starting density	217	-
Vijay clinical	60-day	other sets quoted	isolates with a starting density	210	-7
Vijay clinical	15-day	concentration-duration family	all isolates with an inhibitory concentration	193	-
Vijay clinical	60-day	concentration-duration family	all isolates with an inhibitory concentration	187	-
Vijay clinical	15-day	concentration-duration family	INH-susceptible with one	119	-
Vijay clinical	60-day	concentration-duration family	INH-susceptible with one	117	-
Vijay clinical	15-day	concentration-duration family	baseline only with one	162	-
Vijay clinical	60-day	concentration-duration family	baseline only with one	156	-
ERA4TB six-laboratory	-	-	readings in the file	2775	-
ERA4TB six-laboratory	-	-	excluding inoculum controls	2751	-24
ERA4TB six-laboratory	-	-	flasks (institute x arm x replicate)	90	-
ERA4TB six-laboratory	-	-	treated flasks	72	-18

Deposit	Panel	Stratum	Stage	n	Excluded here
ERA4TB six-laboratory	-	-	treated flasks with a usable starting density	67	-5
ERA4TB six-laboratory	-	-	laboratory-by-arm cells	30	-
ERA4TB six-laboratory	-	-	cells with a fitted kill rate	29	-1
ERA4TB six-laboratory	-	-	series under the plating key (flasks x 4 platings)	360	-
ERA4TB six-laboratory	-	-	treated series	288	-72
ERA4TB six-laboratory	-	-	cross-laboratory pairs in the same arm	191	-
ERA4TB six-laboratory	-	derived units	treated series with a quantified day-0 or day-1 reading in their own plating	262	-26
ERA4TB six-laboratory	-	derived units	of those, not already below their own floor at day zero (the descriptive Cox set)	261	-1
ERA4TB six-laboratory	-	derived units	cross-laboratory pairs, strict rate separation	100	-
ERA4TB six-laboratory	-	derived units	flasks contributing those pairs	42	-
ERA4TB six-laboratory	-	derived units	laboratory-by-arm-by-volume cells with a measured start	56	-
ERA4TB six-laboratory	-	derived units	treated series with three or more quantified readings at 100 uL	64	-
ERA4TB six-laboratory	-	derived units	flask units in the variance decomposition (all arms plus inoculum)	85	-
ERA4TB six-laboratory	-	derived units	untreated day-zero readings at 100 uL (4 laboratories x 3 flasks)	12	-
ERA4TB six-laboratory	-	derived units	below-limit flags in the analysis set	498	-
ERA4TB six-laboratory	-	derived units	readings in the kill-rate analysis set	2580	-171
Dubey hollow fibre (held out)	-	-	cultures with a measured day-zero density	20	-

Table S3. The deposited tolerance class at day 5 after 15 days of prior culture is a threshold on the recorded surviving fraction, with no overlap between classes: low below 10^{-3}, medium from 10^{-3} to 10^{-2} inclusive, high above 10^{-2}. Applying those cuts reproduces every usable class in the file. This matters because the argument that follows is about the fraction the assay recorded, not about an independent clinical judgement.

Recorded class	n	Min fraction	Max fraction	Predicted from cuts	Disagreements
Low	33	3.8×10^{-6}	4.7×10^{-4}	Low	0
Medium	124	1.0×10^{-3}	1.0×10^{-2}	Medium	0
High	46	2.7×10^{-2}	1.0×10^{1}	High	0
All usable	203	-	-	-	0

Table S4. The family of 8 tests between the deposited tolerance label and its candidate determinants, fitted as proportional-odds ordinal logistic regression and corrected together by the Benjamini-Hochberg procedure at a false discovery rate of 5%; the "survives BH" column is that correction. 2 survive. An odds ratio above one means higher odds of a higher tolerance class; time to an optical density (OD) of 0.4 is in days and runs inversely to growth rate, so above one there means *slower* growth accompanies a higher class. The proportional-odds column is a Brant test per predictor; the assumption holds for every predictor in this family. Starting density enters the adjusted fits as a covariate rather than as a member of the family, and it is where proportional odds fails, at 60 days at the deepest endpoint; that failure and the released fit are reported in Section 4. The final column repeats each test on the baseline isolates, one per patient by construction, which is where the resistance association stops clearing its corrected threshold. The deposit carries no patient identifier, so no standard error here can be clustered on the true grouping and every interval in this table is model-based; the baseline-isolate column is the sensitivity analysis that stands in for clustering, and Table S12 records what each conclusion is worth once it is applied.

Predictor	Prior culture	Reading day	For starting density	n	Odds ratio (95% CI)	p	Survives BH	Prop. odc
time to OD 0.4	15 d	day 5	adjusted	202	1.096 (1.03-1.16)	0.0030	yes	0.
resistance	15 d	day 5	unadjusted	202	2.317 (1.30-4.12)	0.0042	yes	0.
time to OD 0.4	60 d	day 2	adjusted	196	0.933 (0.87-1.00)	0.0470	no	0.
time to OD 0.4	15 d	day 2	adjusted	202	1.068 (1.00-1.14)	0.0618	no	0.
resistance	15 d	day 2	unadjusted	202	1.295 (0.66-2.54)	0.4529	no	0.
time to OD 0.4	60 d	day 5	adjusted	196	0.976 (0.91-1.04)	0.4605	no	0.
resistance	60 d	day 2	unadjusted	196	0.836 (0.47-1.48)	0.5400	no	0.
resistance	60 d	day 5	unadjusted	196	1.111 (0.63-1.95)	0.7150	no	0.

Table S5. Decomposition of the isoniazid-resistance association with the tolerance class into a path through log10 starting density and a direct path, by the product of coefficients with bootstrap percentile intervals. The mediated path excludes zero in every specification and the direct path covers zero in every one. This replaces the percentage attenuation the earlier analysis quoted, which is a descriptive ratio rather than an estimand. Sequential ignorability is assumed and is not testable here; the sensitivity analysis in the Methods reports the residual correlation that would nullify the estimate.

Outcome and stratum	n	Total effect c (95% CI)	Direct effect c' (95% CI)	Mediated effect (95% CI)	Proportion mediat
Linear 0/1/2, all IS/IR	203	+0.256 (+0.087, +0.425)	+0.091 (-0.116, +0.287)	+0.166 (+0.067, +0.279)	6
Linear 0/1/2, baseline isolates	168	+0.226 (+0.038, +0.409)	+0.067 (-0.157, +0.294)	+0.159 (+0.053, +0.295)	7
High versus rest	203	+0.121	+0.013	+0.108 (+0.037, +0.189)	8
Not-low versus low	203	+0.135	+0.077	+0.058 (+0.003, +0.121)	4

Table S6. Moxifloxacin at ten times MIC. Every laboratory yields a rate; three record no crossing below the assay floor in any arm at the 100 uL plating. The final column counts first observed crossings, which are not clearances: across the deposit 60% of the series that cross read above the floor again at a later visit. The two censoring estimators agree to 0.003 log10 per day. The starting density is the mean of quantified readings at day 0 or day 1 at the 100 uL plating, pooled over all of that laboratory's arms; the reading-day column says which visits contributed. Institute A deposits no day-zero reading at all, and among treated flasks only institutes C and D do, so for the rest this figure rests partly on readings taken after 24 hours of drug. Text S1 and Table S13 instead use untreated day-zero readings only, so the two tables are not expected to match.

Lab	Starting density (log10 CFU/mL)	Reading day	Kill rate (log10/day)	95% profile interval	By imputation	Readings
A	2.95	day 1	0.119	0.084 to 0.160	0.125	
B	3.16	day 0, day 1	0.139	0.097 to 0.191	0.140	
C	4.02	day 0, day 1	0.464	0.408 to 0.528	0.464	
D	4.69	day 0, day 1	0.104	0.077 to 0.132	0.104	
E	4.95	day 0, day 1	0.090	0.073 to 0.107	0.090	
F	6.53	day 0, day 1	0.223	0.191 to 0.254	0.223	

Table S7. What follows a first observed crossing below the assay floor. The state below the floor is not absorbing: a series sitting below it reads above again at the next visit with probability 0.22 overall, and that probability rises as drug pressure falls, which is what a plating artefact does. Only 56 of 360 series, 15.6 per cent, show the shape a survival model assumes -- one crossing that holds. The 360 series are four platings of each of

90 flasks and are not independent; the flask-level counts are 53 flasks with a crossing series and 42 with a returning one.

Series	n	Visit pairs	P(above to below)	P(below to above)	Never below	One crossing, holds	One crossing, re
all series	360	2232	0.080	0.222	220	56	
INH 10X MIC	72	441	0.137	0.065	34	27	
INH 1X MIC	72	440	0.150	0.373	30	2	
MXF 10X MIC	72	433	0.141	0.175	30	26	
MXF 1X MIC	72	441	0.023	0.923	59	1	
untreated	72	477	0.011	1.000	67	0	

Table S8. Dubey et al. 2026, analysed cold. The floor is derived from a stated 100 uL plated volume and corroborated inside the file: all 229 genuine counts are multiples of ten and the smallest is exactly ten. Starting densities are the 20 measured day-zero counts, not the nominal inoculum the Methods state.

Endpoint	N_reach (per mL)	Cultures	Unreachable	Per cent
1 log (90%)	100	20	0	0%
2 log (99%)	1,000	20	0	0%
3 log (99.9%)	10,000	20	0	0%
4 log (99.99%)	100,000	20	0	0%
5 log (99.999%)	1,000,000	20	5	25%
6 log (99.9999%)	10,000,000	20	20	100%

Table S9. The same 32-fold concentration range summarised at each sampling day (upper rows), and the concentration slope fitted separately in each interval (lower rows). Slopes and intervals are from the replicate-level bootstrap described in the Methods.

Read at	Survivor ratio (low/high dose)	log10 separation	Slope per doubling	95% interval	p
day 3	6.9x	0.84			
day 7	48.2x	1.68			
day 14	5.2x	0.71			
days 0-3			+0.0590	+0.0327 to +0.0834	0.013
days 3-7			+0.0330	+0.0079 to +0.0586	0.236
days 7-14			-0.0251	-0.0353 to -0.0153	0.091

Table S10. The deepest reduction each culture in the prospective experiment could report, at each of its two platings. $h = \log 10(N0/L)$ with L one colony in the pooled volume plated, taken at the lowest dilution the series was read at, which is where the floor is lowest and the reportable depth greatest. The two platings of one flask share a row: at the standard inoculum the 10 µL plating ceiling runs 3.71 to 3.93 logs and the 100 µL ceiling 4.88 to 5.05, so a four-log endpoint is unreportable at one plating and reportable at the other in the same culture. The gain column is close to the log10(10) = 1.00 that plating ten times the volume buys; it is not exactly 1.00 because each plating measures its own $N0$ and the two measurements differ.

Arm	Flask	N_0 at 10 µL (per mL)	N_0 at 100 µL (per mL)	h at 10 µL	h at 100 µL	Gain
High	1	3,600,000	4,050,000	4.86	5.91	+1.05
High	2	4,450,000	3,300,000	4.95	5.82	+0.87
High	3	4,450,000	3,950,000	4.95	5.90	+0.95
Low	1	58,000	78,000	3.06	4.19	+1.13
Low	2	42,500	65,000	2.93	4.11	+1.18
Low	3	45,500	50,000	2.96	4.00	+1.04
Mid	1	425,000	565,000	3.93	5.05	+1.12
Mid	2	340,000	380,000	3.83	4.88	+1.05
Mid	3	255,000	375,000	3.71	4.88	+1.17

Table S11. How often one culture receives two different tolerance labels from its two platings, swept across the class threshold. That two platings report different fractions is arithmetic; that those fractions land either side of a cut is not, and this is the quantity that could have come out zero. At a cut of one per cent the window is nearly shut. At one in a thousand, where the clinical classification reanalysed here cuts its lowest class, it is one sample-time in nine. The two pairs of columns differ in one thing. The first holds the starting density to a single value per flask, so the only thing separating the two readings is the plated volume, which is what the argument claims. The second lets each plating carry the starting density it measured for itself, which is what a laboratory running both platings would have; the count rises because those two estimates of one culture disagree by 0.74- to 1.53-fold. The first column is the claim, the second is the practice, and neither is the other.

Class threshold c_1	Straddling, one N_0 per flask	Share	Straddling, each plating's own N_0	Share
10^-2	1 of 54	2%	1 of 54	2%
10^-3	6 of 54	11%	9 of 54	17%
10^-4	7 of 54	13%	8 of 54	15%

Table S12. Every conclusion this paper draws from the two primary deposits, against uncertainty recomputed at the level the observations are actually independent. 20 survive unchanged, 6 survive with materially wider uncertainty, 5 do not survive, and 1 is withdrawn because its null is false before any data are seen. Each verdict applies to the claim as it was originally stated. Three of the five failures are gone from the text entirely; for the other two a weaker statement is retained and is marked as such where it appears -- the Cox coefficient trade is now reported as hazard ratios with no p-value, and institute C as the laboratory whose crossings its starting density does not account for rather than as a tested contrast. The five that fail are all between-laboratory p-values computed on flasks: every flask in a laboratory shares a starting culture, so a comparison that looks like 67 flasks is six laboratories, and for a three-against-three split of six clusters the smallest attainable two-sided p is 0.10. They are deleted rather than corrected, because there is nothing to correct them to.

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead
4	the difference between panels in the growth association is itself supported (interaction p = 0.0072)	unknown isolate-panel (each of 210 isolates counted once per panel)	210 isolates	not recomputable
5	starting density separates flasks that ever crossed below the assay floor from those that never did (AUC 0.974, p = 1.2e-10)	67 flask	6	exact laboratory-level test (3 crossed vs 3 did not); cluster bootstrap over laboratories; permutation of crossing within laboratory, and within laboratory AND treatment arm
5	in a Cox model, institutes D, E and F carry p = 0.015, 0.016, 0.015, which adjustment for starting density moves to 0.138, 0.172, 0.576	67 flask	6	cluster-robust sandwich on six clusters
5	the log-rank test separates the six laboratories on time to first crossing	67 flask	6	none available: the grouping variable is the cluster
5	institute C remains distinguishable after adjustment (HR 5.27, p < 0.001)	67 flask	1	none available
1	all 33 short isolates are at the ceiling; 88.0% of the rest are (Fisher p = 0.030)	217 isolate	174 patients min	none: the null is arithmetically impossible
5	at ten times MIC of moxifloxacin the kill rate spans 5.2-fold between laboratories (0.090 to 0.464)	6 laboratory-arm cell	6	cluster bootstrap over whole laboratories
5	at one times MIC five of six laboratories record net growth	6 laboratory-arm cell	6	Jeffreys interval on six clusters
5 / 7	the laboratory effect lands on the duration endpoint and not on the rate; the rate is 'far more reproducible'	42 flask	6	permutation of the laboratory label across flasks, within arm for the rate
6	the deepest demonstrable kill differs by 2.33 log10 between laboratories at one plating volume	12 day-zero reading	4	cluster bootstrap over whole laboratories

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead
7	34.6% of 191 cross-laboratory pairs are inversions (95% CI 28.1-41.5)	191 pair of flasks	6	cluster bootstrap over whole laboratories; also over flasks within laboratory; also delete-one-laboratory jackknife
7	under the strictest rate separation the inversion rate is 23.0% of 100 pairs (95% CI 15.6-31.9)	100 pair of flasks	6	cluster bootstrap over whole laboratories
1	33 of 217 isolates (15.2%) lack the headroom for a 4-log endpoint	217 isolate	174 patients min	baseline-only recount
1 / 2	18 isolates rest on the floor at 15 days and 6 at 60	217 + 210 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	exact McNemar on the paired binary
10	no MIC-MDK association survives Benjamini-Hochberg (supplementary text, not a numbered section of the article)	6 tests isolate	174 patients min	baseline-only family of six
2	18 isolates at the floor; their survival span IS their inoculum span	217 isolate	174 patients min	baseline-only recount
2	185 determinable, 12 forced by the inoculum, 6 undecidable	203 isolate	174 patients min	baseline-only reclassification
2	at 60 days the observability counts are 191, 6 and none, 'because the cultures are denser and few readings reach the floor'	203 + 197 isolate-panel (each of 210 isolates counted once per panel)	197 isolates	exact McNemar on the paired binary
2 / 4	the deep endpoint is unreachable for 26.2% of resistant against 7.6% of susceptible isolates	217 isolate	174 patients min	baseline-only Fisher exact
4	resistant isolates enter the assay ten-fold lower	217 isolate	174 patients min	baseline-only Mann-Whitney
4	the resistance-tolerance association attenuates once starting density enters the model, and its interval then spans zero	202 isolate	174 patients min	baseline-only refit

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead
4	growth state predicts the tolerance class and survives adjustment (beta = +0.027, p = 0.0025)	202 isolate	174 patients min	baseline-only refit
4	resistant isolates do not grow measurably more slowly (p = 0.24)	217 isolate	174 patients min	baseline-only Mann-Whitney
4	the confound thins between panels: 15.2% short of headroom at 15 days against 3.3% at 60	217 + 210 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	exact McNemar on the paired binary
4	the 60-day cultures are denser, so headroom is larger there	420 treated as independent isolate-panel (each of 210 isolates counted once per panel)	210 isolates	Wilcoxon signed-rank on the within-isolate change
4	the spread of starting densities more than halves between panels (IQR 1.00 to 0.42 log10)	420 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	paired bootstrap resampling whole isolates
5	of 64 treated series the final step is not a decline in 48 (75%)	64 series (one per treated flask)	6	cluster bootstrap over whole laboratories
5	44 of 64 series end more than one log10 above their own nadir	64 series (one per treated flask)	6	cluster bootstrap over whole laboratories
5 / Limitations	87.0% of the variance in flask starting density lies between laboratories	85 flask	6	permutation of the laboratory label across flasks
6	including the choice of plated volume widens the spread in measurable depth to 4.40 log10	up to 4 volumes x 5 laboratories laboratory-by-volume channel	6	no resampling required for the volume term
7	the criterion D_A/D_B > b_A/b_B calls 83.8% of pairs correctly	191 pair of flasks	6	cluster bootstrap over whole laboratories
Methods	83 of 498 below-limit flags (16.7%) are contradicted by another plating of the same sample at the same visit	498 flag (one reading)	6	cluster bootstrap over whole laboratories

Table S13. The deepest reduction each assay could resolve, as h = log10(N0/L), with the assay floor taken per sample where it varies. Rows compare only within a level of variation: the clinical rows describe between-isolate starting burden, which is biological, and are not a measure of laboratory imprecision. Kaur is NA because its three day-zero readings are technical replicates of one preparation and cannot estimate between-preparation reproducibility, and because that deposit states no quantification limit.

Dataset	Level of variation	n	Median log10 N0	Assay floor L	delta h (log10)	Fold
ERA4TB, between laboratories at 100 uL	between laboratories, one written protocol	12	4.63	10 CFU/mL, derived from 100 uL plated	2.33	216x
ERA4TB, every laboratory and plating volume	laboratories and plated volumes together	56	4.29	10, 100 or 400 CFU/mL by volume	4.40	25,123x
Vijay, 15-day culture	between clinical isolates	217	5.79	23 MPN/mL, inferred	4.42	26,522x
Vijay, 60-day culture	between clinical isolates	210	7.36	23 MPN/mL, inferred	5.43	269,565x
Kaur, planktonic	technical replicates of one preparation	3	7.03	not stated in the deposit	NA	NA
Kaur, intracellular	technical replicates of one preparation	3	5.94	not stated in the deposit	NA	NA
Dubey, hollow fibre	between cultures, one laboratory	20	6.08	10 CFU/mL, derived from 100 uL plated	0.60	4x

Table S14. Fold below the nominal 0.5 McFarland reference, 1.5e8 CFU/mL. Descriptive only, and not a protocol-compliance metric. A time-kill inoculum is prepared by diluting from a suspension matched to that turbidity, so every entry is expected to sit far below it; the conversion of a turbidity to CFU/mL depends on species, cell aggregation and preparation and is least reliable for mycobacteria. The clinical rows are most probable numbers divided by a reference stated in CFU/mL, so those two ratios cross units and are the least meaningful in the table.

Dataset	Median log10 N0	Fold below nominal 0.5 McFarland
ERA4TB, between laboratories at 100 uL	4.63	3,525x
ERA4TB, every laboratory and plating volume	4.29	7,751x
Vijay, 15-day culture	5.79	246x
Vijay, 60-day culture	7.36	7x
Kaur, planktonic	7.03	14x
Kaur, intracellular	5.94	170x
Dubey, hollow fibre	6.08	123x

Table S15. Pairs of flasks in the same arm from different laboratories. An inversion is a pair in which the population that fell faster crossed below the assay floor later. 45 further pairs that the censoring could not settle are excluded rather than imputed. The variance columns decompose the spread in crossing time. The rate term is the larger at the flask level in all three arms that can be decomposed, and that ordering is NOT a claim this table supports: the flasks are three to a laboratory and share a starting culture, so the quantity is a between-laboratory statistic on six clusters. Deleting one laboratory sends the distance share above a half in two of the three arms, and a bootstrap over whole laboratories straddles a half in all three. The last two columns are those two checks, and they are the reason the text claims only that the two terms are of comparable size. Moxifloxacin at one times MIC is not among them: the decomposition is taken on log D and log b, five of six laboratories record net growth in that arm, and only one flask in it returns a positive fitted rate, so there is no spread in the rate to divide.

Arm	Comparable pairs	Inversions	Rate	95% CI	Called by D/b criterion	Variance: distance	Variance: rate	Dis
All arms pooled	191	66	34.6%	28.1-41.5%	84.3%			
INH 10X MIC						32.3%	67.7%	
INH 1X MIC						17.7%	82.3%	
MXF 10X MIC						35.2%	64.8%	

Table S16. The six strongest of the 24 comparisons the 217-isolate file supports. 4 reach nominal significance where 1.2 are expected by chance; none exceeds its Benjamini-Hochberg critical value, which is ranked against the full family of 24 rather than against any one stratum. MDK99 and MDK99.99 are the minimum durations for a 99 and a 99.99 per cent reduction, the endpoints the text names in words. The final column is the correlation each design could have resolved at 95% confidence.

Stratum	Endpoint	n	Spearman rho	p	BH critical value	Survives correction	Resolvable rho
baseline only	MDK99.99 (15d)	162	-0.206	0.0086	0.0021	no	0.15
INH-susceptible	MDK99.99 (15d)	119	-0.218	0.0171	0.0042	no	0.18
INH-susceptible	MDK99 (60d)	117	-0.198	0.0324	0.0063	no	0.18
all isolates	MDK99.99 (60d)	187	-0.153	0.0361	0.0083	no	0.14
INH-susceptible	MDK99.99 (60d)	117	-0.179	0.0540	0.0104	no	0.18
baseline only	MDK99.99 (60d)	156	-0.149	0.0628	0.0125	no	0.16

Table S17. The same nominal concentration expressed in multiples of the minimum inhibitory concentration each population actually evolved to. At 25 ug/mL the same number denotes a sub-inhibitory exposure in one nutrient condition and a strongly inhibitory one in another.

Nominal concentration (ug/mL)	Nutrient levels	Lowest exposure (x MIC)	Highest exposure (x MIC)	Spread	Straddles t
12.5	3	1.49	4.74	3.2x	
25	3	0.55	9.47	17.1x	
50	2	5.08	10.95	2.2x	
100	1	9.72	9.72	1.0x	

Table S18. Isoniazid-resistant isolates enter this assay ten-fold lower than susceptible ones, and we do not know why. This is what the deposit can and cannot rule out. The verdict column is decided on the two recorded-rate columns beside it: PROXY means the covariate is recorded for no isolate in one of the two exposure groups, so its missingness is the exposure; PARTIAL means both groups record it but the missingness is still tied to the exposure (Fisher exact p < 10^{-6}); INDEPENDENT means it is not. Seven of the nine covariates are therefore unusable, and they fail in two ways. Five are recorded almost exclusively for resistant isolates: the two susceptibility calls, the two Mykrobe calls and the mutation identity. The susceptibility calls are missing on identical rows, which is why they return identical coefficients, and the Mykrobe calls carry a single value wherever they are recorded in this stratum, so no model can be fitted for them and no coefficient is printed. The other two are the drug MICs, recorded for every susceptible isolate and 67 of 84 resistant ones; the isoniazid MIC separates the two groups completely, so adjusting for it conditions on a graded reading of the exposure. A negative attenuation is an amplification: the adjusted coefficient sits further from zero than the unadjusted one, which is what the isoniazid MIC does at 13 per cent. A coefficient is printed wherever one exists so the circularity is visible, but none of these seven bounds anything. Only 2 covariates are recorded at rates unrelated to susceptibility, and adjusting for either leaves the coefficient within five per cent of its unadjusted value. The file records no referring site and no processing batch, so those cannot be tested at all.

Covariate	Recorded, resistant	Recorded, susceptible	Can it adjust?	Resistance coefficient	Attenuation
INH_MGIT_DST	82/84	0/119	PROXY	-0.851	-0%
RIF_MGIT_DST	82/84	0/119	PROXY	-0.851	-0%
INH_mutation	79/84	1/119	PARTIAL	-0.662	22%
INH_Mykrobe	76/84	0/119	PROXY	-	-
RIF_Mykrobe	76/84	0/119	PROXY	-	-
MIC_INH	67/84	119/119	PARTIAL	-0.963	-13%
MIC_RIF	67/84	119/119	PARTIAL	-0.853	-0%
Time_to_0.4	83/84	119/119	INDEPENDENT	-0.807	5%
Time_point	84/84	119/119	INDEPENDENT	-0.818	4%

Table S19. What the assay floor *L* is in each deposit analysed. No deposit reports a validated limit of quantification with a value. The six-laboratory file names the concept in the definitions of its below- and above-quantification-limit columns but defines it as whether the plate was countable, which is an operator's judgement. DERIVED means the value follows arithmetically from a recorded plated volume; INFERRED means it was read off the deposit's own behaviour and is labelled as inferred wherever it is used; NONE means no floor is evidenced and none is assumed; FLAGGED means the deposit marks below-limit readings without naming a value, which fixes no floor either. Where a plated volume is recorded the honest term is the minimum reportable positive count, one colony in that volume; elsewhere it is an operational assay floor.

Deposit	States an LOD	States an LOQ	Value used	Units	How obtained	What it should be called
Vijay 2024	no	no	23	MPN per mL	INFERRED	operational assay floor
ERA4TB 2023	no	term only	10, 100 and 400, per plated volume	CFU per mL	DERIVED	minimum reportable positive count
Windels 2024	no	no	none; no value is recoverable	surviving fraction, dimensionless	FLAGGED	no floor is established; a below-limit reading is written as exact zero and fixes no value
Kaur 2024	no	no	none; no floor is used	log10 CFU per mL	NONE	no floor is evidenced; the smallest reading is the smallest reading
Dubey 2026	no	no	10	CFU per mL	DERIVED	minimum reportable positive count

Table S20. Sensitivity of the load-bearing counts to the choice of floor. Each deposit is sensitive to a different thing, so the table is long rather than wide: one row per deposit, floor scenario and quantity. The clinical sweep steps through every three-tube most-probable-number rung at or below 23 per mL, and the number of isolates short of four logs of headroom moves only between 28 and 33 across the whole range, which is why the inferred floor is safe to use. The six-laboratory rows price what pooling the four plating volumes to one floor would cost. The Kaur deposit records no plated volume, so its rows show what assuming one would do: the four-log endpoint stays reachable throughout while the headroom itself moves by 1.6 log10.

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 3 MPN/mL	isolates short of 4 logs	28
Vijay 2024	floor = 3 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 3 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 3 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 3 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 3 MPN/mL	short of 4 logs, baseline only	17
Vijay 2024	floor = 3.6 MPN/mL	isolates short of 4 logs	28
Vijay 2024	floor = 3.6 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 3.6 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 3.6 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 3.6 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 3.6 MPN/mL	short of 4 logs, baseline only	17
Vijay 2024	floor = 7.2 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 7.2 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 7.2 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 7.2 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 7.2 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 7.2 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 7.4 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 7.4 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 7.4 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 7.4 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 7.4 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 7.4 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 9.2 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 9.2 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 9.2 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 9.2 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 9.2 MPN/mL	calls with several compatible classes	0

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 9.2 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 11 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 11 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 11 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 11 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 11 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 11 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 14 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 14 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 14 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 14 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 14 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 14 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 15 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 15 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 15 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 15 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 15 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 15 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 20 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 20 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 20 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 20 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 20 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 20 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 21 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 21 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 21 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 21 MPN/mL	calls with one compatible class	0

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 21 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 21 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 23 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 23 MPN/mL	isolates at the floor	18
Vijay 2024	floor = 23 MPN/mL	calls resting on a measured fraction	185
Vijay 2024	floor = 23 MPN/mL	calls with one compatible class	12
Vijay 2024	floor = 23 MPN/mL	calls with several compatible classes	6
Vijay 2024	floor = 23 MPN/mL	short of 4 logs, baseline only	21
ERA4TB	per-volume (as analysed)	genuine counts a pooled floor discards (%)	0
ERA4TB	per-volume (as analysed)	below-limit flags contradicted (%)	16.67
ERA4TB	pooled at the most sensitive volume, 100 uL	genuine counts a pooled floor discards (%)	0
ERA4TB	pooled at the most sensitive volume, 100 uL	below-limit flags contradicted (%)	43.17
ERA4TB	pooled at the modal volume, 10 uL	genuine counts a pooled floor discards (%)	1.13
ERA4TB	pooled at the modal volume, 10 uL	below-limit flags contradicted (%)	28.71
ERA4TB	pooled at the least sensitive volume, 2.5 uL	genuine counts a pooled floor discards (%)	5.94
ERA4TB	pooled at the least sensitive volume, 2.5 uL	below-limit flags contradicted (%)	15.26
ERA4TB	pooled at the geometric mean of the per-volume floors	genuine counts a pooled floor discards (%)	1.04
ERA4TB	pooled at the geometric mean of the per-volume floors	below-limit flags contradicted (%)	38.96
Dubey 2026	the below-limit placeholder taken literally	median headroom (log10)	6.08
Dubey 2026	the below-limit placeholder taken literally	cultures short of 4 logs	0
Dubey 2026	100 uL plated, as the deposit's staging record gives	median headroom (log10)	5.08
Dubey 2026	100 uL plated, as the deposit's staging record gives	cultures short of 4 logs	0
Dubey 2026	50 uL plated	median headroom (log10)	4.78
Dubey 2026	50 uL plated	cultures short of 4 logs	0
Dubey 2026	10 uL plated	median headroom (log10)	4.08
Dubey 2026	10 uL plated	cultures short of 4 logs	5
Kaur 2024	100 uL plated, if it had been stated	headroom (log10)	6.03
Kaur 2024	100 uL plated, if it had been stated	4-log endpoint reachable	True
Kaur 2024	the observed minimum, which occurs once	headroom (log10)	5.43

Deposit	Floor assumed	Quantity	Value
Kaur 2024	the observed minimum, which occurs once	4-log endpoint reachable	True
Kaur 2024	10 uL plated, if it had been stated	headroom (log10)	5.03
Kaur 2024	10 uL plated, if it had been stated	4-log endpoint reachable	True
Kaur 2024	2.5 uL plated, if it had been stated	headroom (log10)	4.43
Kaur 2024	2.5 uL plated, if it had been stated	4-log endpoint reachable	True
Windels 2024	no floor is recoverable	readings written as exact zero	6

Table S21. The comparison the Methods promise: rows dropped from the association family against rows retained, on starting density and susceptibility. The MDR exclusion differs in susceptibility by construction, since those isolates are outside the resistant-versus-susceptible contrast the family tests. The one difference not by construction is in the 60-day panel, where the dropped rows sit higher in starting density (p = 0.027); it affects 20 rows and no conclusion drawn from that panel.

Panel	Exclusion	Dropped	Retained	Median log10 N0 retained	Median log10 N0 dropped	p	Resistant, retainec
15-day	label missing or 'MDR'	14	202	5.79	5.36	0.198	41%
15-day	growth proxy missing	1	202	5.79	6.79	0.149	41%
60-day	label missing or 'MDR'	20	196	7.36	7.79	0.027	40%
60-day	growth proxy missing	1	196	7.36	7.36	0.749	40%

Table S22. The ordinal reanalysis against the linear model it replaces, member for member. The linear model scores the ordering 0, 1, 2, which assumes the two class steps are equal; the ordinal model does not. Every direction agrees and the same two members survive correction under both, so the linear treatment did not manufacture the result -- but the coefficients it reports are in a unit that does not exist, which is why the ordinal fit is the one in the main table.

Predictor	Prior culture	Reading day	Linear beta	p (linear)	Survives BH	Odds ratio	p (ordinal)	Survives BH	Sa
time to OD 0.4	15 d	day 5	+0.0266	0.0025	yes	1.096	0.0030	yes	
resistance	15 d	day 5	+0.2589	0.0035	yes	2.317	0.0042	yes	
time to OD 0.4	60 d	day 2	-0.0195	0.0590	no	0.933	0.0470	no	
time to OD 0.4	15 d	day 2	+0.0116	0.0812	no	1.068	0.0618	no	
resistance	15 d	day 2	+0.0515	0.4446	no	1.295	0.4529	no	
time to OD 0.4	60 d	day 5	-0.0041	0.6999	no	0.976	0.4605	no	
resistance	60 d	day 2	-0.0551	0.5486	no	0.836	0.5400	no	
resistance	60 d	day 5	+0.0389	0.6797	no	1.111	0.7150	no	

References cited in this supplemental material

1. National Committee for Clinical Laboratory Standards. 1999. Methods for determining bactericidal activity of antimicrobial agents; approved guideline. NCCLS document M26-A. National Committee for Clinical Laboratory Standards, Wayne, PA. ISBN 1-56238-384-1.

2. McFarland J. 1907. The nephelometer: an instrument for estimating the number of bacteria in suspensions used for calculating the opsonic index and for vaccines. JAMA 49:1176. https://doi.org/10.1001/jama.1907.25320140022001f.

3. Franzblau SG, DeGroote MA, Cho SH, Andries K, Nuermberger E, Orme IM, Mdluli K, Angulo-Barturen I, Dick T, Dartois V, Lenaerts AJ. 2012. Comprehensive analysis of methods used for the evaluation of compounds against *Mycobacterium tuberculosis*. Tuberculosis (Edinb) 92:453-488. https://doi.org/10.1016/j.tube.2012.07.003.

4. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*. Nat Commun 17:7504. https://doi.org/10.1038/s41467-026-74323-2.

5. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Source Data to "Molecular pharmacodynamics of amoxicillin-clavulanic acid

for urinary tract infections caused by *Escherichia coli*" (41467_2026_74323_MOESM4_ESM.xlsx). Nature Communications. https://doi.org/10.1038/s41467-026-74323-2.

6. Brauner A, Fridman O, Gefen O, Balaban NQ. 2016. Distinguishing between resistance, tolerance and persistence to antibiotic treatment. Nat Rev Microbiol 14:320-330. https://doi.org/10.1038/nrmicro.2016.34.

7. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence. ISME J 18:wrae070. https://doi.org/10.1093/ismejo/wrae070.

8. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Data supporting "Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence". Zenodo. https://doi.org/10.5281/zenodo.7550302.

9. Pym AS, Saint-Joanis B, Cole ST. 2002. Effect of *katG* mutations on the virulence of *Mycobacterium tuberculosis* and the implication for transmission in humans. Infect Immun 70:4955-4960. https://doi.org/10.1128/IAI.70.9.4955-4960.2002.

10. van Soolingen D, de Haas PEW, van Doorn HR, Kuijper E, Rinder H, Borgdorff MW. 2000. Mutations at amino acid position 315 of the *katG* gene are associated with high-level resistance to isoniazid, other drug resistance, and successful transmission of *Mycobacterium tuberculosis* in the Netherlands. J Infect Dis 182:1788-1790. https://doi.org/10.1086/317598.

11. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Supplementary file 2 to "Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study" (elife-93243-supp2-v1.xlsx). eLife. https://doi.org/10.7554/eLife.93243.

12. van Wijk RC, Lucía A, Sudhakar PK, Sonnenkalb L, Gaudin C, Hoffmann E, Dremierre B, Aguilar-Ayala DA, Dal Molin M, Rybniker J, de Giorgi S, Cioetto-Mazzabò L, Segafreddo G, Manganelli R, Degiacomi G, Recchia D, Pasca MR, Simonsson USH, Ramón-García S. 2023. Implementing best practices on data generation and reporting of *Mycobacterium tuberculosis* in vitro assays within the ERA4TB consortium. iScience 26:106411. https://doi.org/10.1016/j.isci.2023.106411.

13. van Wijk RC, Lucía Quintana A, Ramón-García S. 2023. *Mycobacterium tuberculosis* time kill assay data of the standardized protocol within the ERA4TB consortium. figshare. https://doi.org/10.6084/m9.figshare.19766083.v1.

14. Rabodoarivelo MS, Hoffmann E, Gaudin C, Aguilar-Ayala DA, Galizia J, Sonnenkalb L, Dal Molin M, Cioetto-Mazzabò L, Degiacomi G, Recchia D, Rybniker J, Manganelli R, Pasca MR, Ramón-García S, Lucía A. 2025. Protocol to quantify bacterial burden in time-kill assays using colony-forming units and

most probable number readouts for *Mycobacterium tuberculosis*. STAR Protoc 6:103643. https://doi.org/10.1016/j.xpro.2025.103643.

15. Kaur P, Ramya VK, Naveenkumar CN, Bharathkumar K, Singh M, Hobbie SN, Shandil RK, Narayanan S. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis*. Front Trop Dis 5:1413211. https://doi.org/10.3389/fitd.2024.1413211.
16. Kaur P. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis* — raw data. figshare. https://doi.org/10.6084/m9.figshare.26462791.v1.
17. Cochran WG. 1950. Estimation of bacterial densities by means of the "most probable number". Biometrics 6:105. https://doi.org/10.2307/3001491.
18. Blodgett R. 2023. BAM appendix 2: most probable number from serial dilutions. Bacteriological analytical manual. US Food and Drug Administration, Silver Spring, MD. https://www.fda.gov/food/laboratory-methods-food/bam-appendix-2-most-probable-number-serial-dilutions.
19. Imai K, Keele L, Yamamoto T. 2010. Identification, inference and sensitivity analysis for causal mediation effects. Stat Sci 25:51-71. https://doi.org/10.1214/10-STS321.
20. Baron RM, Kenny DA. 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. J Pers Soc Psychol 51:1173-1182. https://doi.org/10.1037/0022-3514.51.6.1173.
21. Tobin J. 1958. Estimation of relationships for limited dependent variables. Econometrica 26:24. https://doi.org/10.2307/1907382.
22. Beal SL. 2001. Ways to fit a PK model with some data below the quantification limit. J Pharmacokinet Pharmacodyn 28:481-504. https://doi.org/10.1023/A:1012299115260.
23. Rubin DB. 1987. Multiple imputation for nonresponse in surveys. John Wiley & Sons, New York, NY. https://doi.org/10.1002/9780470316696.
24. McCullagh P. 1980. Regression models for ordinal data. J R Stat Soc Series B Stat Methodol 42:109-127. https://doi.org/10.1111/j.2517-6161.1980.tb01109.x.
25. Brant R. 1990. Assessing proportionality in the proportional odds model for ordinal logistic regression. Biometrics 46:1171-1178.
26. Peterson B, Harrell FE Jr. 1990. Partial proportional odds models for ordinal response variables. J R Stat Soc Ser C Appl Stat 39:205. https://doi.org/10.2307/2347760.
27. Turnbull BW. 1976. The empirical distribution function with arbitrarily grouped, censored and truncated data. J R Stat Soc Series B Stat Methodol 38:290-295. https://doi.org/10.1111/j.2517-6161.1976.tb01597.x.

28. Kaplan EL, Meier P. 1958. Nonparametric estimation from incomplete observations. J Am Stat Assoc 53:457-481. https://doi.org/10.1080/01621459.1958.10501452.

29. Cox DR. 1972. Regression models and life-tables. J R Stat Soc Series B Stat Methodol 34:187-202. https://doi.org/10.1111/j.2517-6161.1972.tb00899.x.

30. Field CA, Welsh AH. 2007. Bootstrapping clustered data. J R Stat Soc Series B Stat Methodol 69:369-390. https://doi.org/10.1111/j.1467-9868.2007.00593.x.

31. Schoenfeld D. 1982. Partial residuals for the proportional hazards regression model. Biometrika 69:239-241. https://doi.org/10.1093/biomet/69.1.239.

32. Grambsch PM, Therneau TM. 1994. Proportional hazards tests and diagnostics based on weighted residuals. Biometrika 81:515-526. https://doi.org/10.1093/biomet/81.3.515.

33. Efron B. 1979. Bootstrap methods: another look at the jackknife. Ann Stat 7:1-26. https://doi.org/10.1214/aos/1176344552.

34. Brown LD, Cai TT, DasGupta A. 2001. Interval estimation for a binomial proportion. Stat Sci 16:101-133. https://doi.org/10.1214/ss/1009213286.

35. Benjamini Y, Hochberg Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Series B Stat Methodol 57:289-300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

36. Harris CR, Millman KJ, van der Walt SJ, Gommers R, Virtanen P, Cournapeau D, Wieser E, Taylor J, Berg S, Smith NJ, Kern R, Picus M, Hoyer S, van Kerkwijk MH, Brett M, Haldane A, del Río JF, Wiebe M, Peterson P, Gérard-Marchant P, Sheppard K, Reddy T, Weckesser W, Abbasi H, Gohlke C, Oliphant TE. 2020. Array programming with NumPy. Nature 585:357-362. https://doi.org/10.1038/s41586-020-2649-2.

37. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones E, Kern R, Larson E, Carey CJ, Polat İ, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 1.0 Contributors. 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods 17:261-272. https://doi.org/10.1038/s41592-019-0686-2.

38. McKinney W. 2010. Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference 56-61. https://doi.org/10.25080/Majora-92bf1922-00a.

39. Seabold S, Perktold J. 2010. Statsmodels: econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference 92-96. https://doi.org/10.25080/Majora-92bf1922-011.

40. Davidson-Pilon C. 2019. lifelines: survival analysis in Python. J Open Source Softw 4:1317. https://doi.org/10.21105/joss.01317.